

4 Image Enhancement

-Frequency domain

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2. The Fourier series & the Fourier transform
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 - Image smoothing
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4. Fast Fourier Transform

Jean Baptiste Joseph Fourier



Fourier was born in Auxerre,
France in 1768

- Most famous for his work “*La Théorie Analitique de la Chaleur*” published in 1822
- Translated into English in 1878:
“*The Analytic Theory of Heat*”

Nobody paid much attention when the work was first published.

One of the most important mathematical theories in modern engineering.

A *Fourier series*: Any function that periodically repeats itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient.

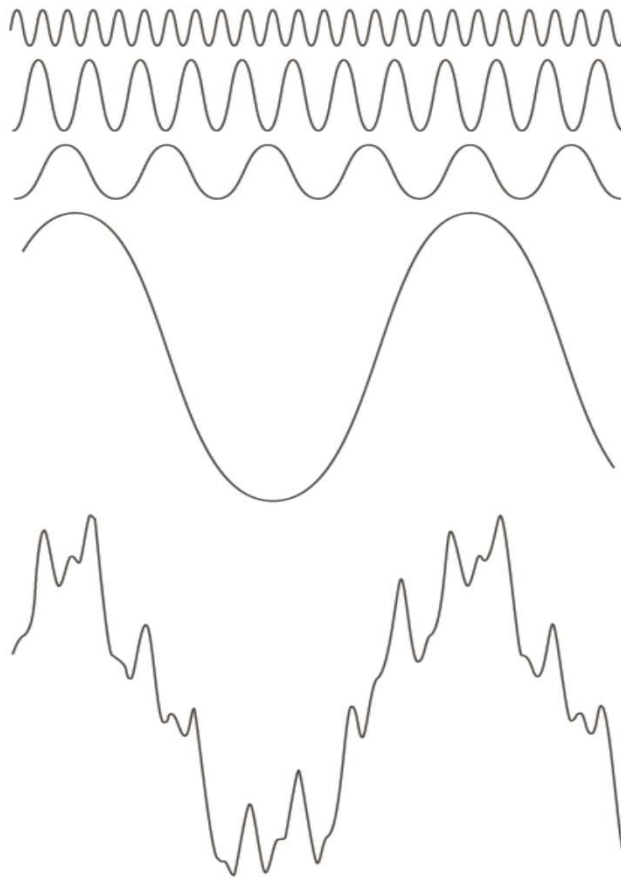
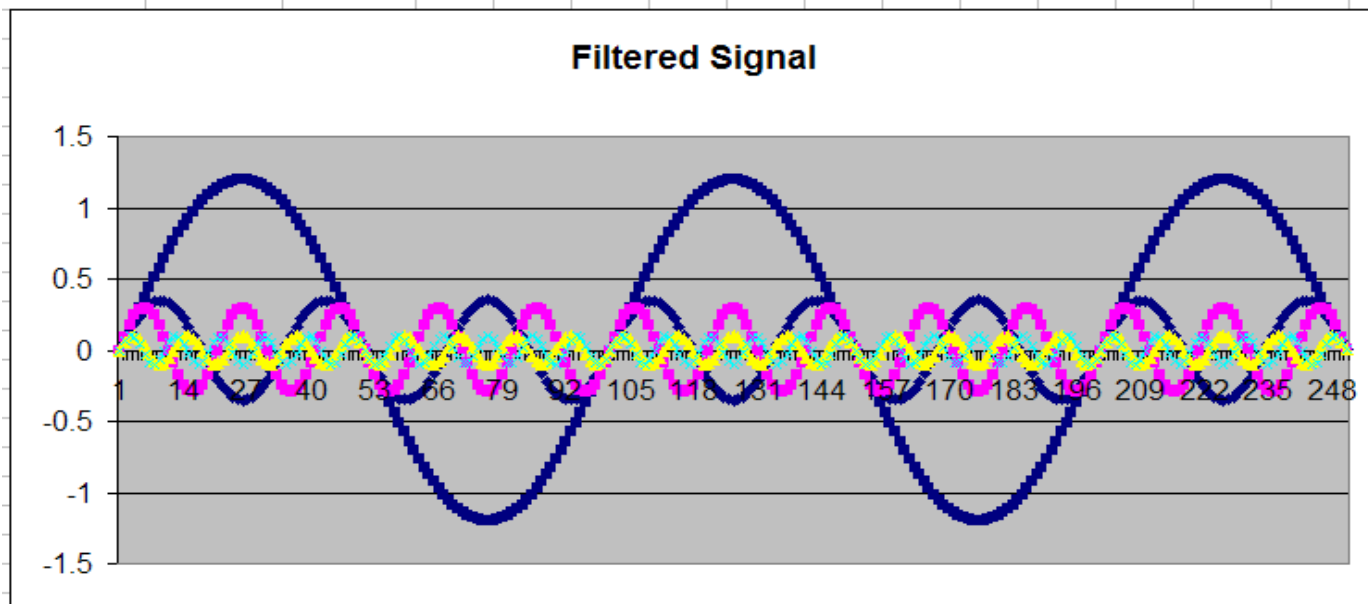
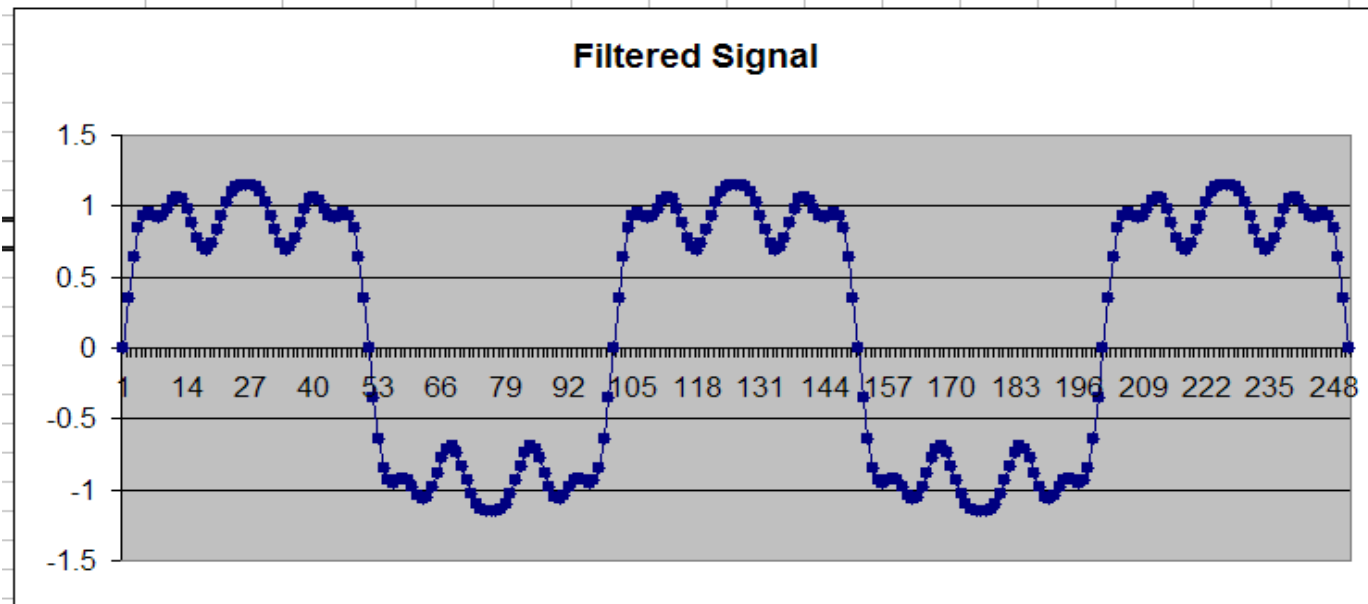


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

Frequency domain signal processing example



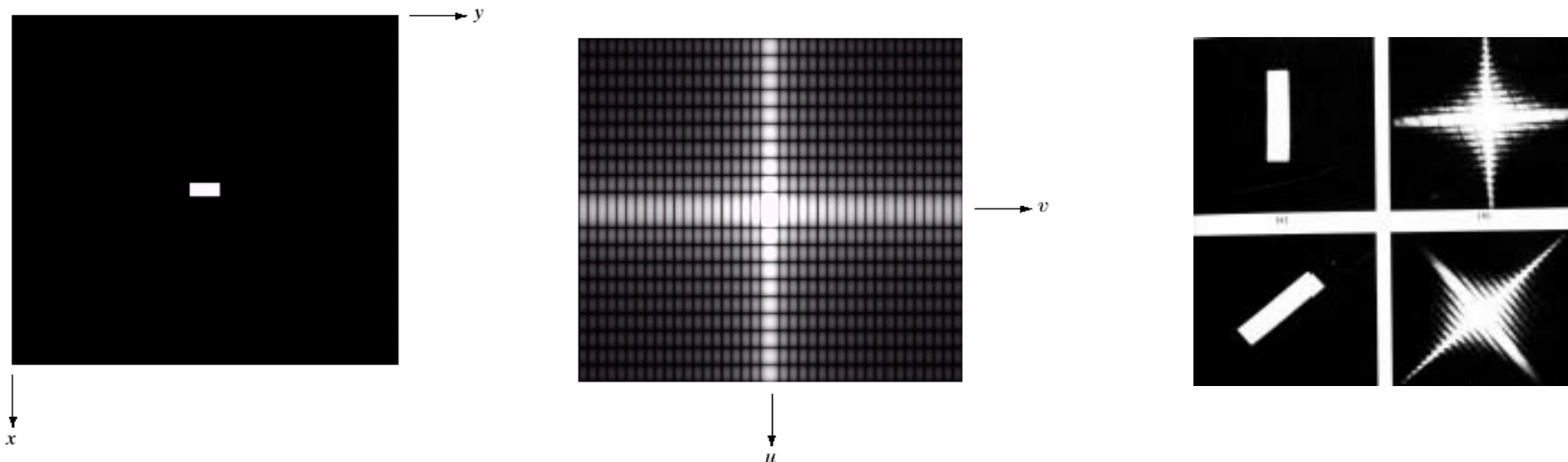
1 2D Discrete Fourier Transform (DFT)

The *Discrete Fourier Transform* of $f(x, y)$, for $x = 0, 1, 2 \dots M-1$ and $y = 0, 1, 2 \dots N-1$, denoted by $F(u, v)$, is given by the equation:

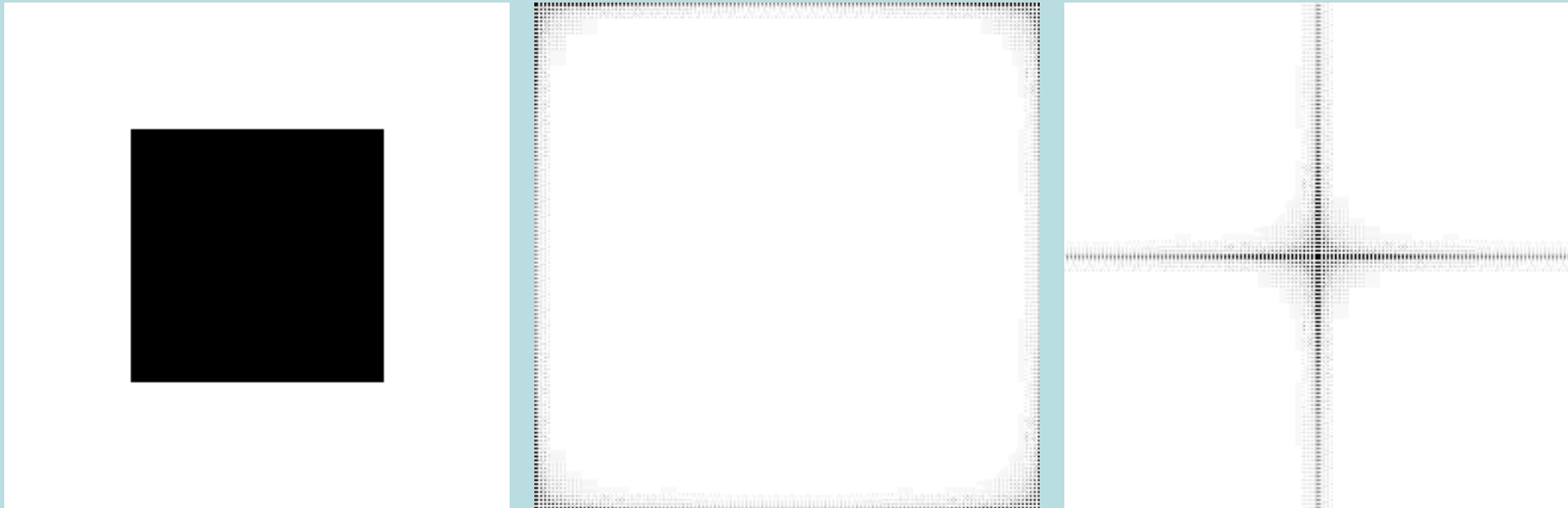
$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

where $u = 0, 1, 2 \dots M-1$ and $v = 0, 1, 2 \dots N-1$.

x and y : spatial variables; u, v : frequency variables



1 2D Discrete Fourier Transform (DFT)



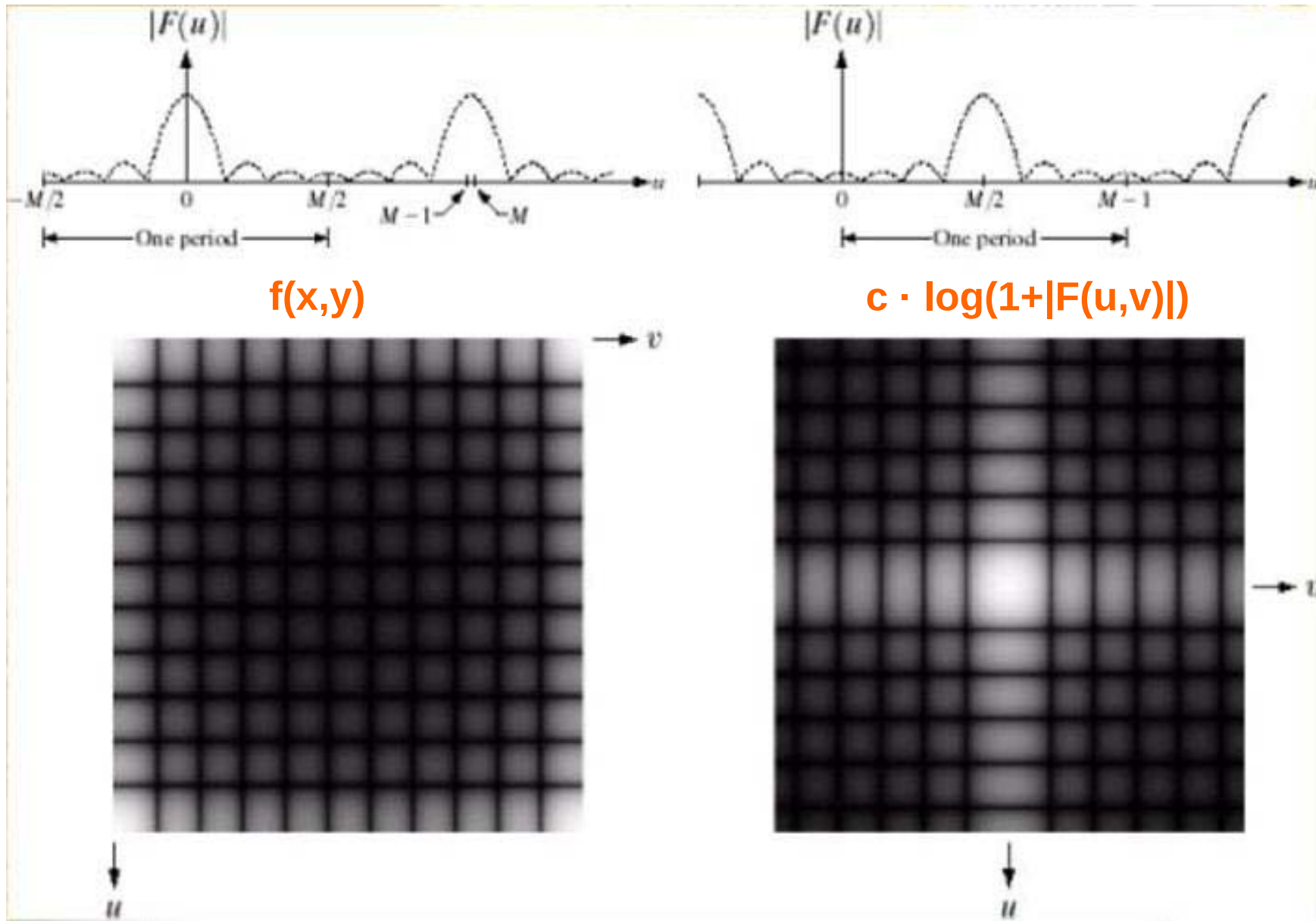
Step 1. Centering the Spectrum helps with:

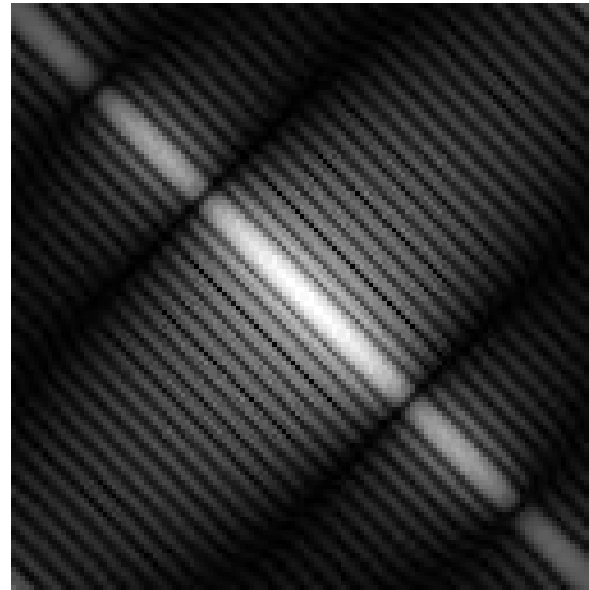
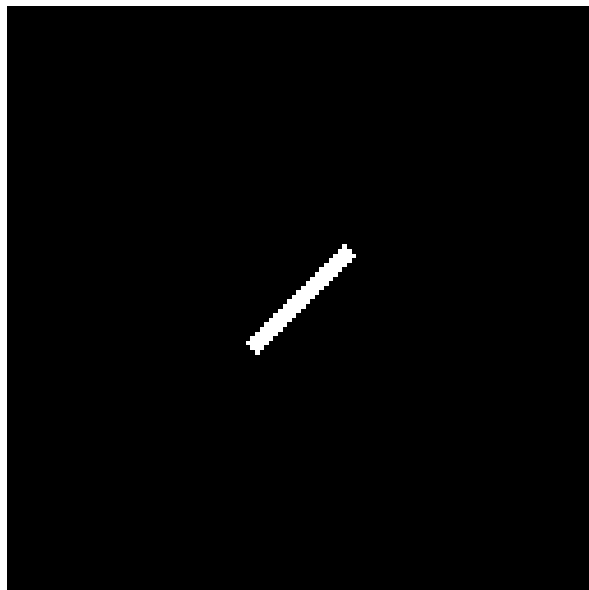
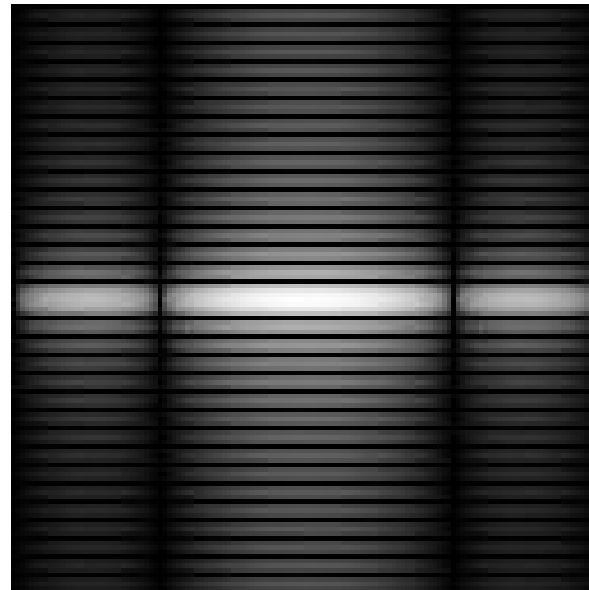
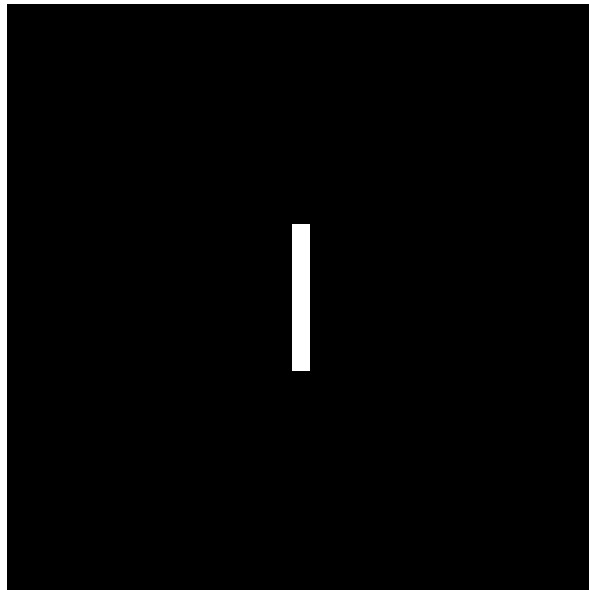
- a) Visualization
- b) Simplifies filtering

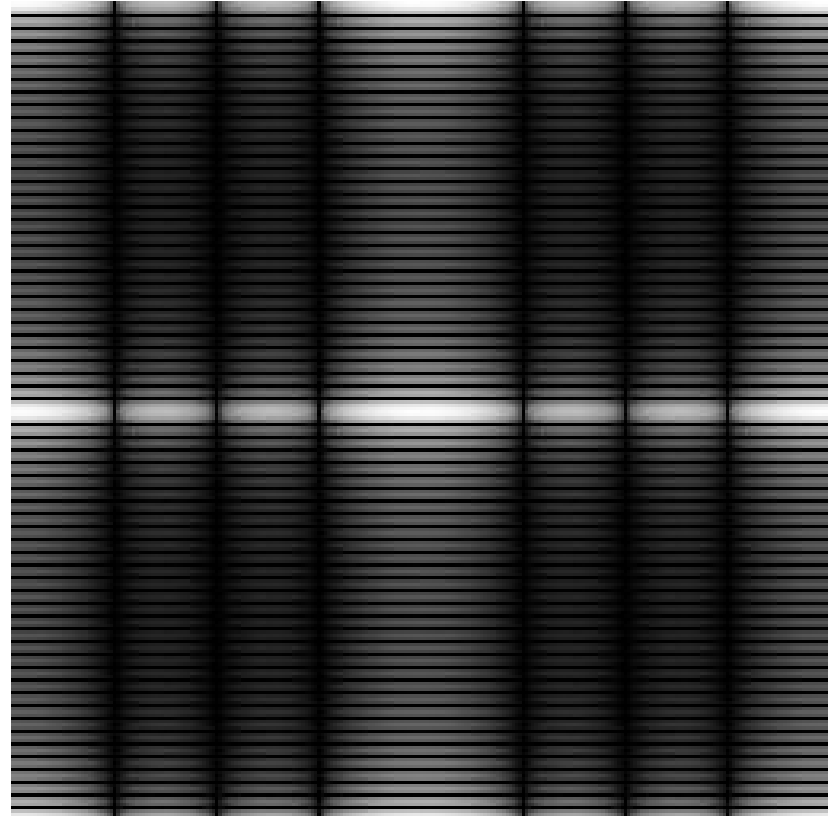
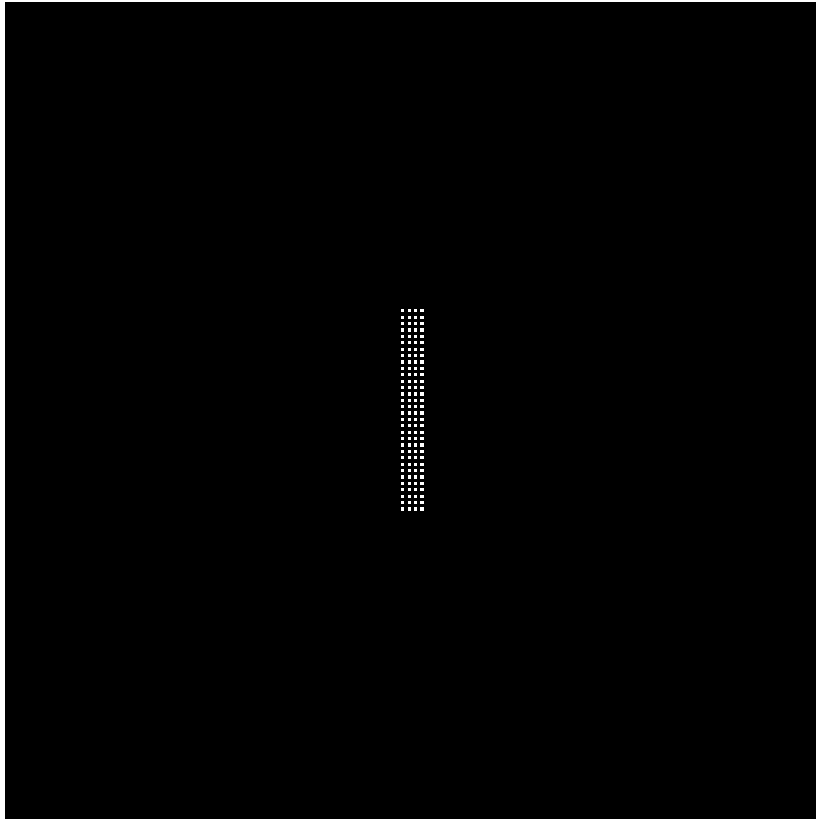
$$f(x,y) \rightarrow f(x,y) \cdot (-1)^{x+y}$$

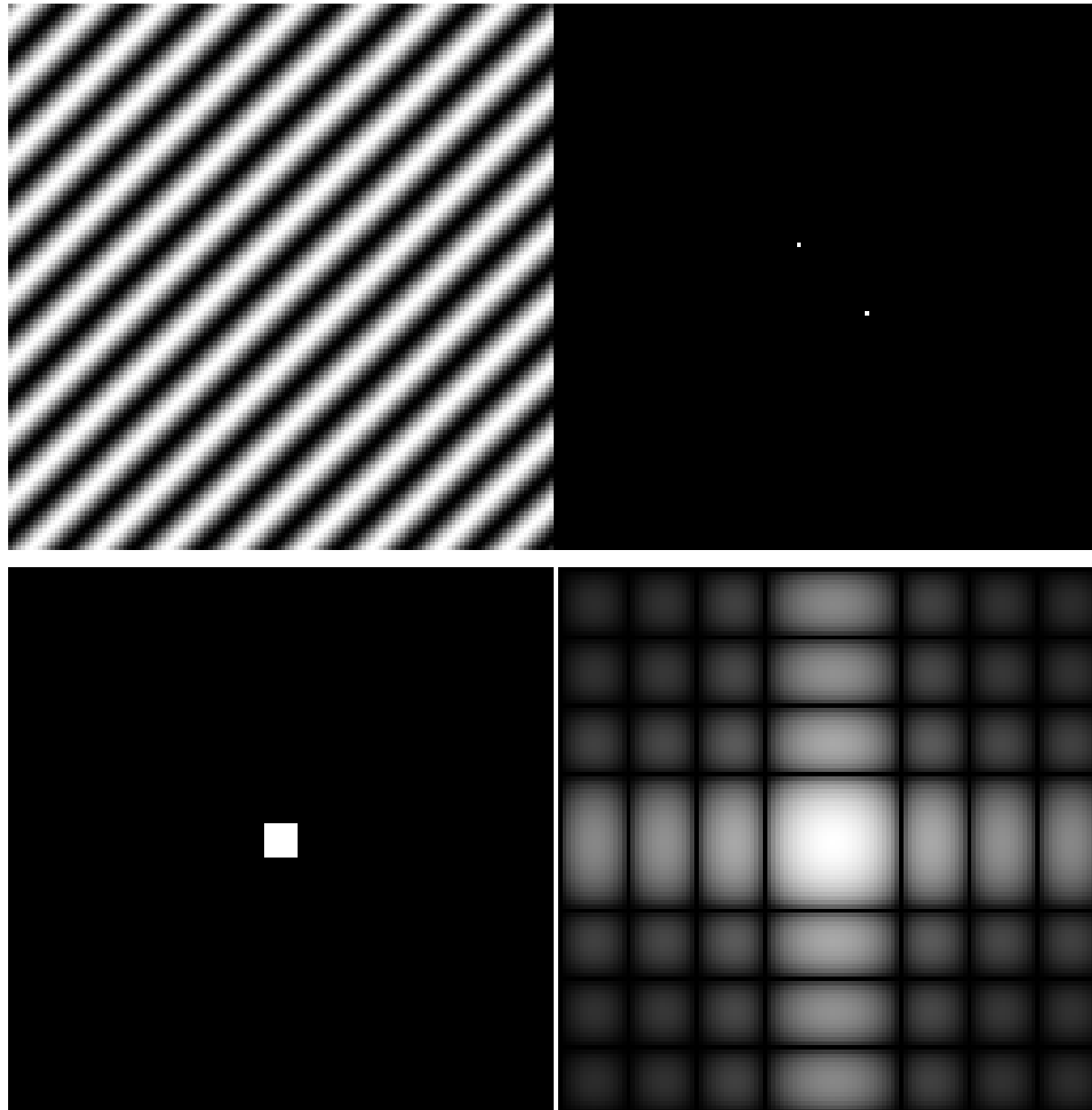
Step 2. Log transformation: compresses large dynamic range.

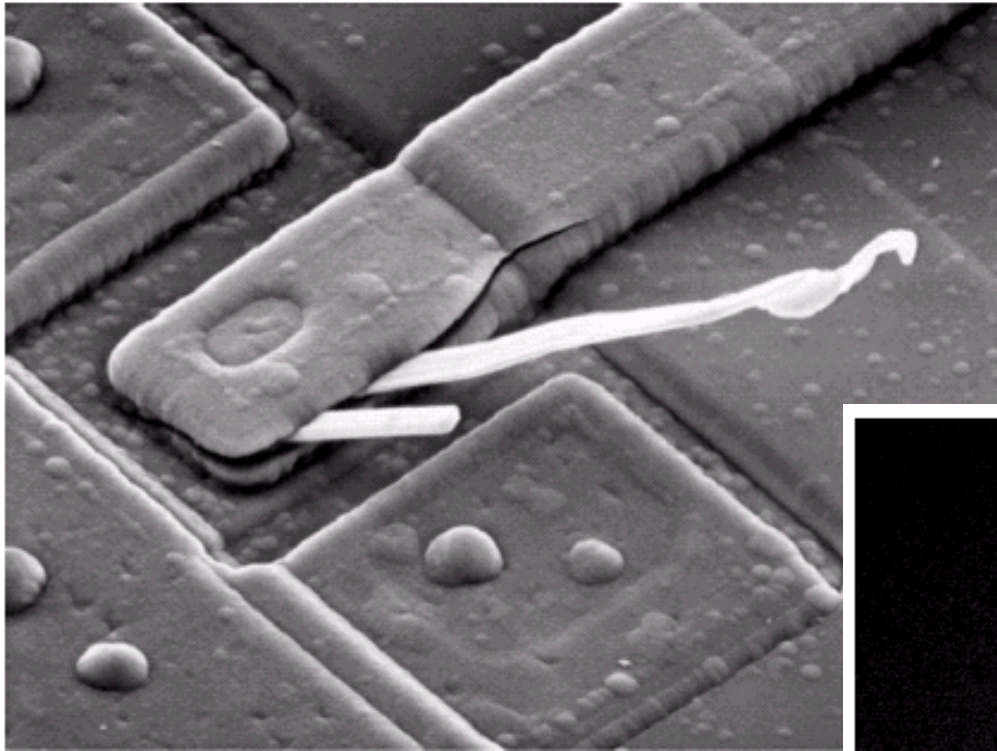
1 2D Discrete Fourier Transform (DFT)



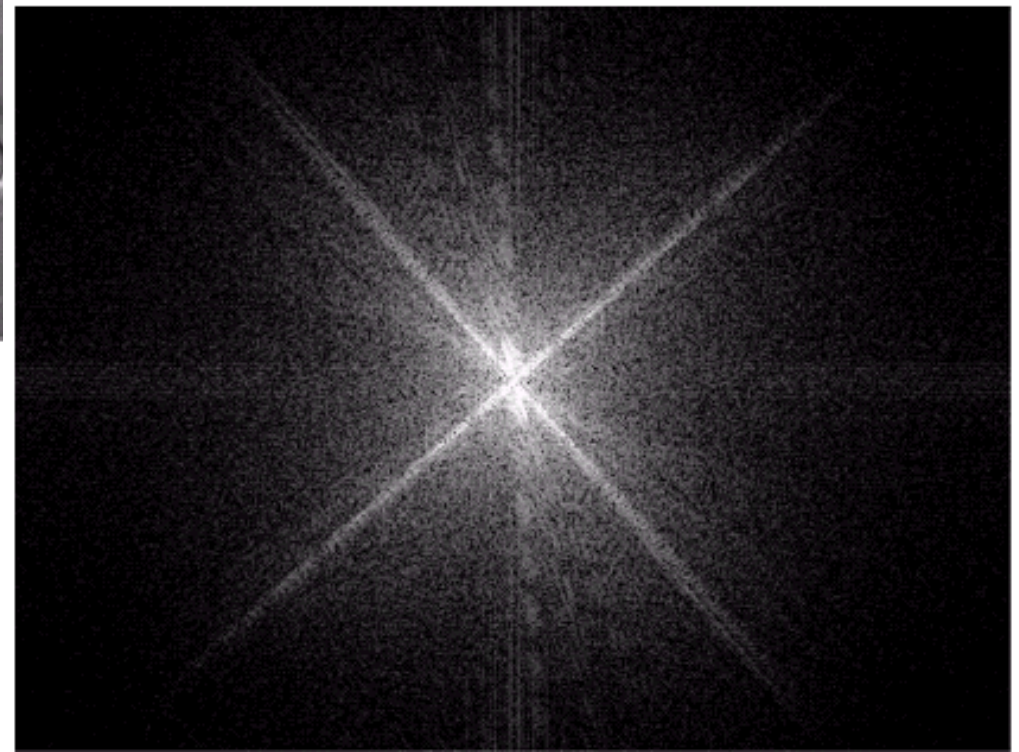




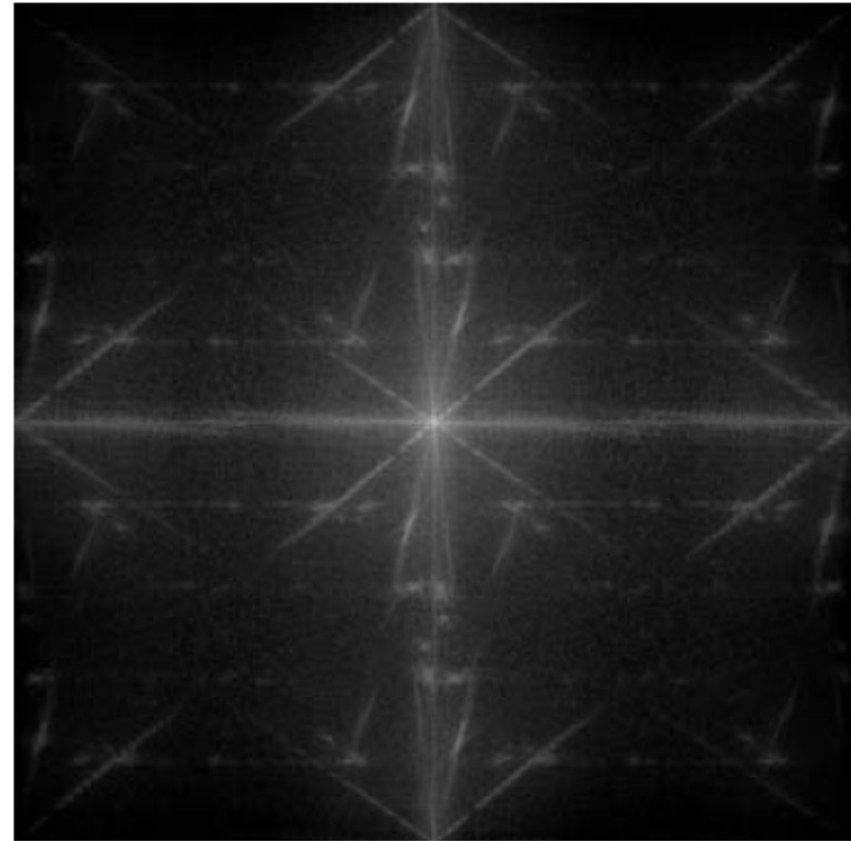




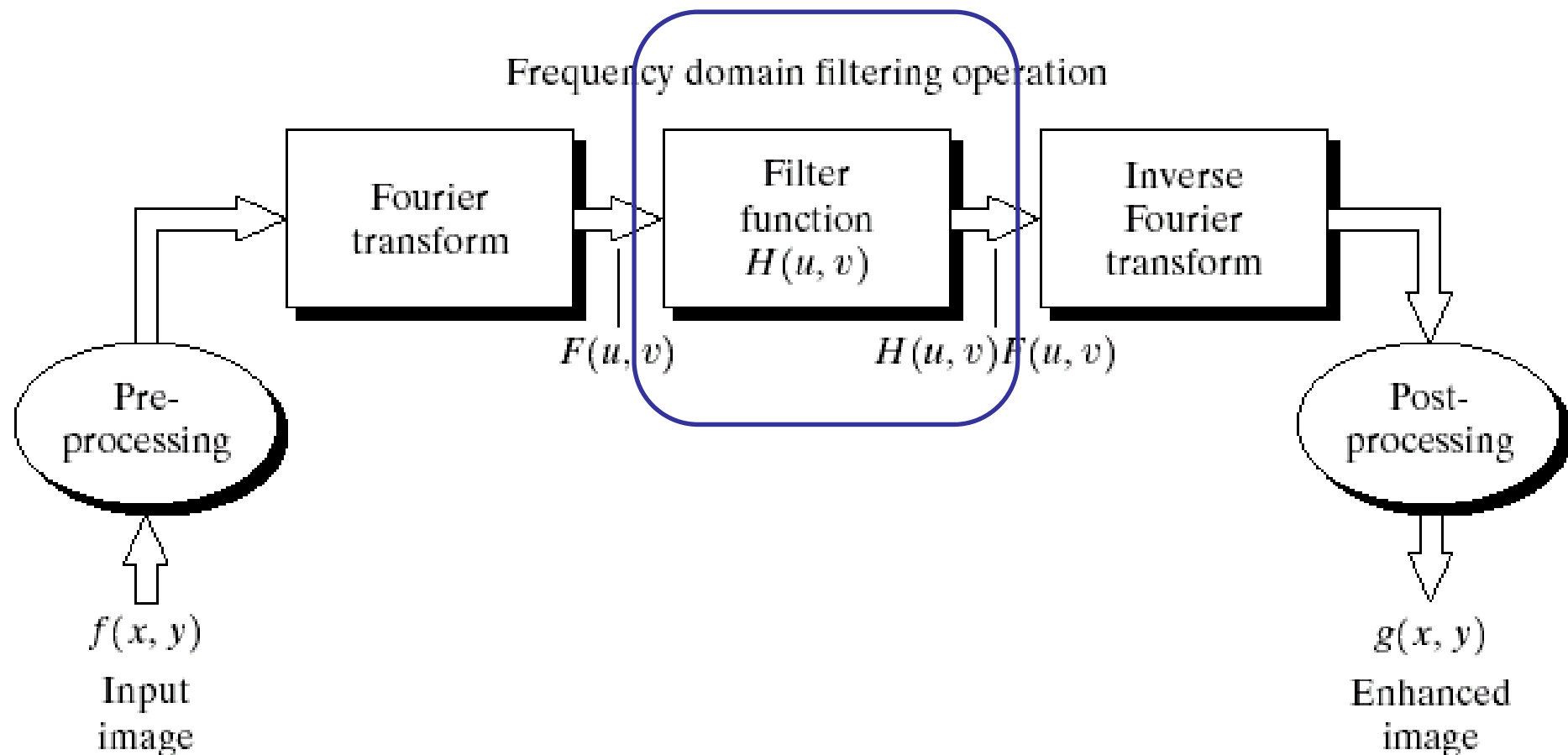
Scanning electron microscope image
of an integrated circuit magnified
~2500 times



Fourier spectrum of the image



2 Inverse 2D DFT



2 Inverse 2D DFT

Fourier transform is completely reversible.

The inverse DFT is given by:

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

for $x = 0, 1, 2 \dots M-1$ and $y = 0, 1, 2 \dots N-1$

Some properties of the 2D DFT

a) $F(0,0)$ = average intensity ("DC component")

$$F(0,0) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

b) if $f(x,y)$ is real $\rightarrow F(u,v) = F^*(-u,-v) \rightarrow |F(u,v)| = |F(-u,-v)|$

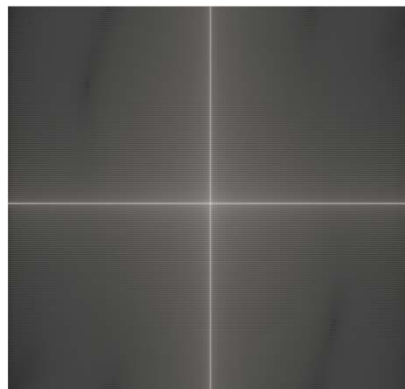
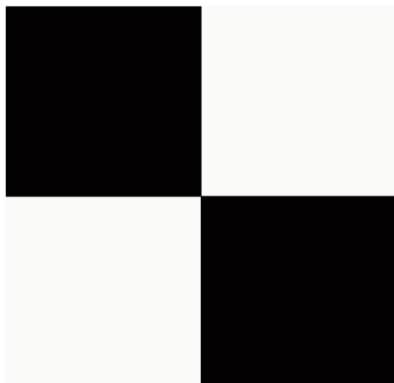
$$\Delta u = \frac{1}{M\Delta x} \quad \Delta v = \frac{1}{N\Delta y}$$

u and v determine frequency and orientation of sinusoids on the x - y plane (spatial frequency components)

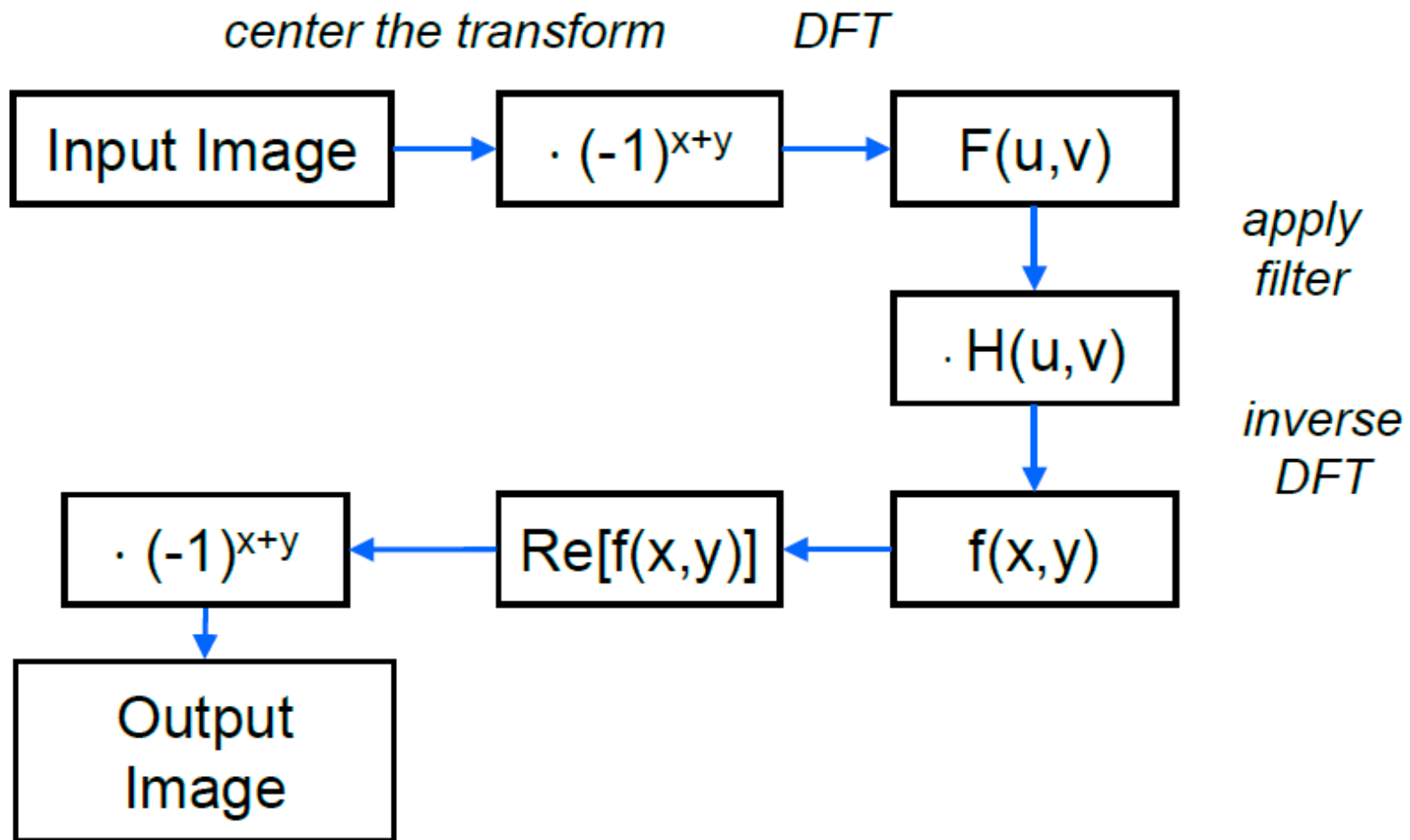
$\Rightarrow$ Fourier transform "measures the amount of a sinusoid with given frequency and orientation in the signal"

Discrete Fourier Transform programming in Matlab

```
f = imread('chessboard_pattern.tif');  
figure; imshow(f);  
F = fft2(f); % Fourier transform of f  
Fc = fftshift(F); % center transform  
specFc = abs(Fc); % spectrum of F  
figure; imshow(log(1 + specFc), []);  
f2 = real(ifft2(F)); % inverse Fourier transform  
figure; imshow(f2, []);
```

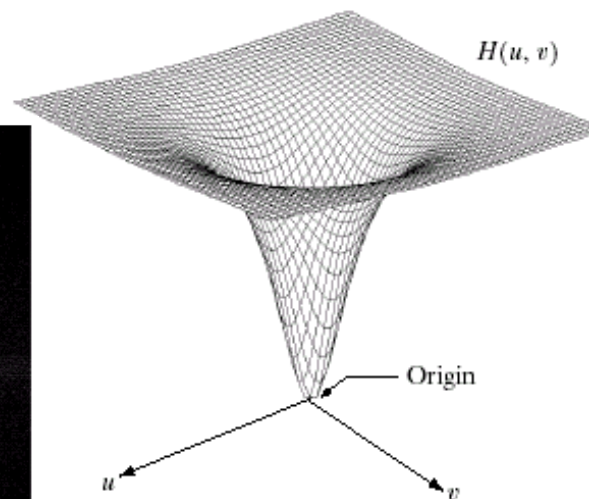
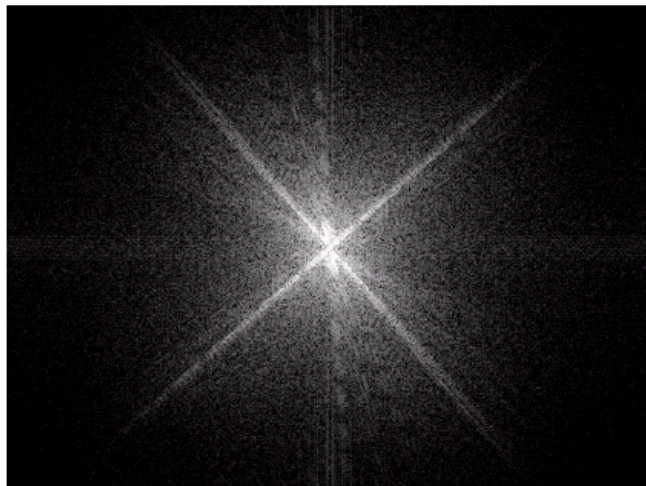
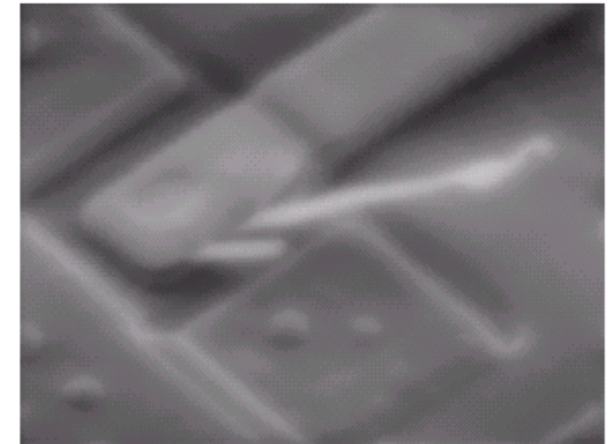
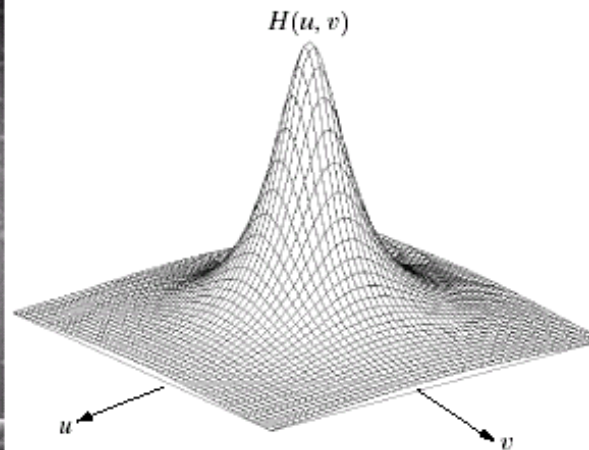
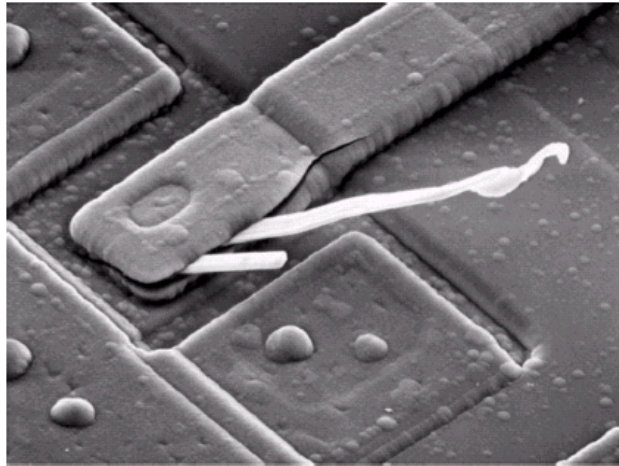


3 Frequency domain filtering

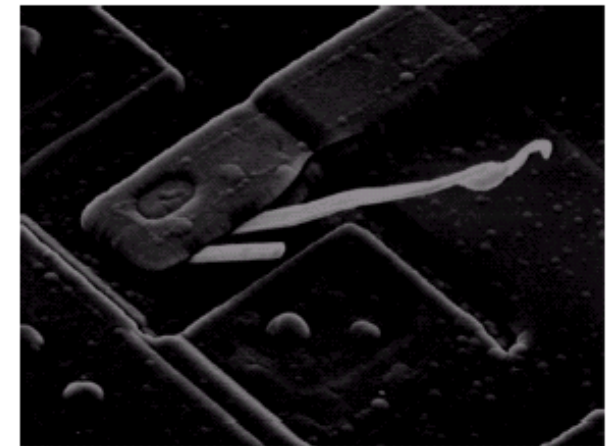


3 Frequency domain filtering

Low Pass Filter



High Pass Filter



Smoothing Frequency Domain Filters—Ideal Low pass filters

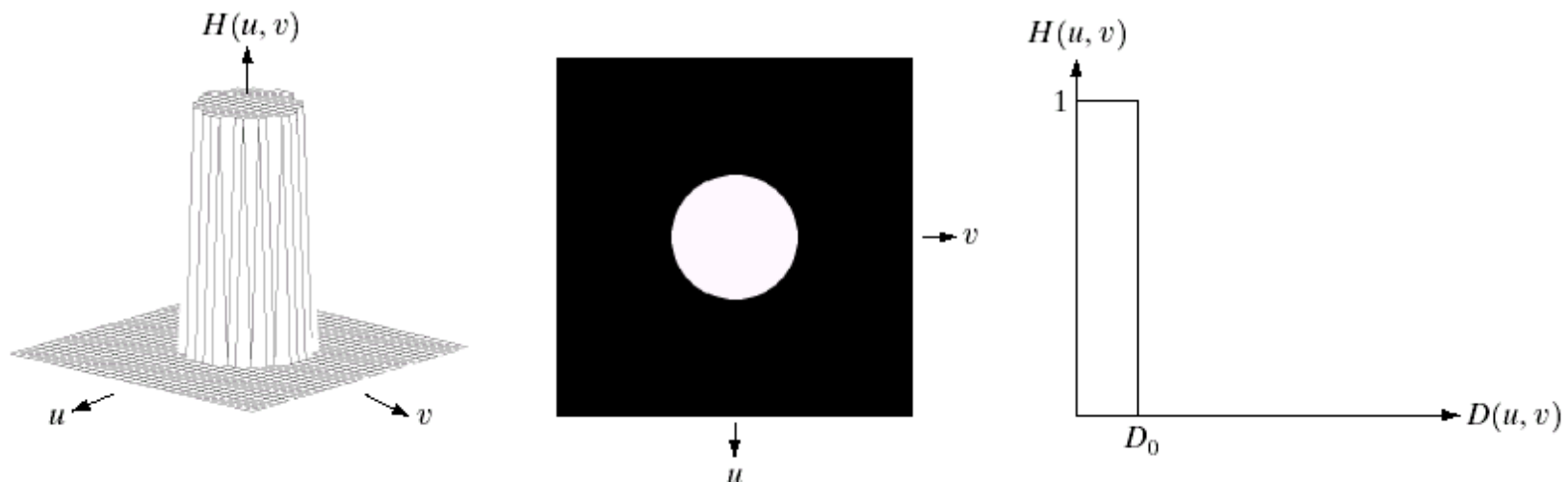
Smoothing in the frequency domain: dropping out the high frequency components (edges, noise).

Basic model for filtering:

$$G(u,v) = H(u,v)F(u,v)$$

where $F(u,v)$ is the Fourier transform of the image, and $H(u,v)$ is the filter transform function.

Low pass filters: pass the low frequencies, drop the high ones.



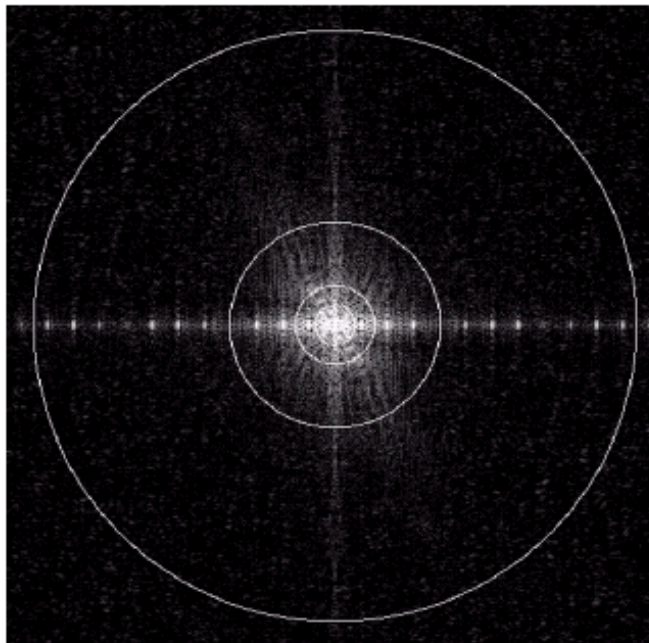
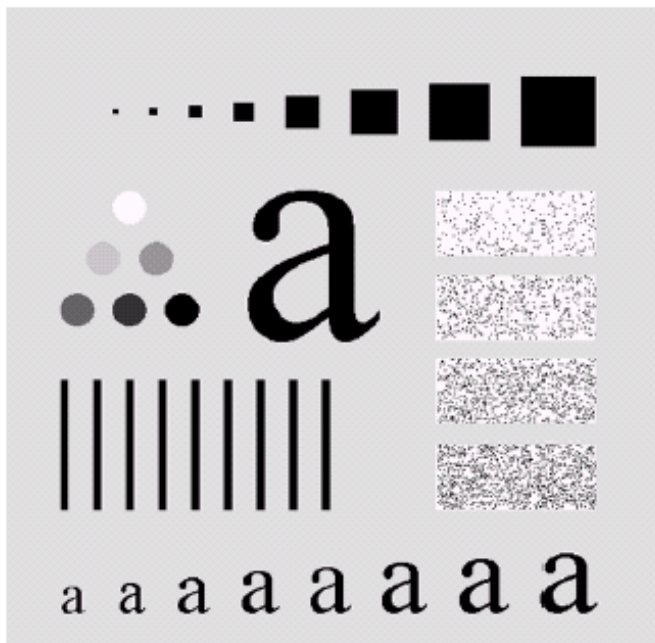
Smoothing Frequency Domain Filters—Ideal Low pass filters

The transfer function for the ideal low pass filter:

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

where $D(u, v)$ is given as (distance to the centre):

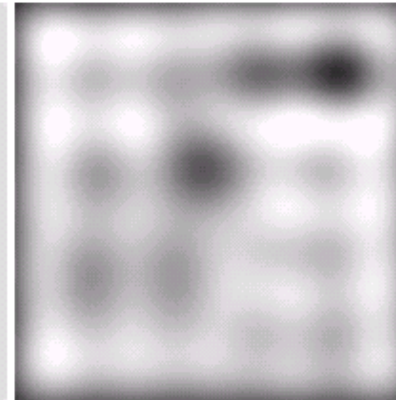
$$D(u, v) = [(u - M / 2)^2 + (v - N / 2)^2]^{1/2}$$



Fourier spectrum
and a series of ideal
low pass filters of
radius 5, 15, 30, 80
and 230
superimposed on
top of it

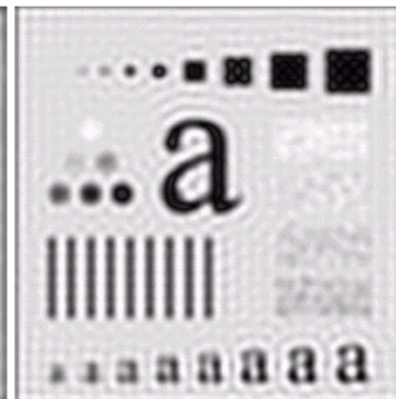
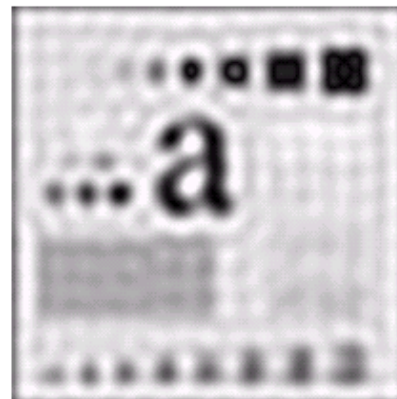
Smoothing Frequency Domain Filters—Ideal Low pass filters

Original image



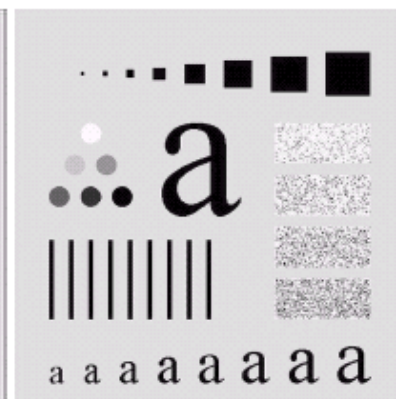
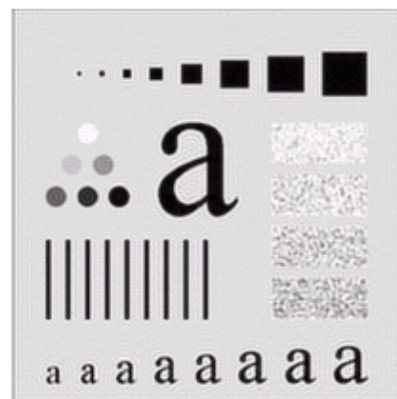
Result of filtering with
ideal low pass filter of
radius 5

Result of filtering
with ideal low pass
filter of radius 15



Result of filtering with
ideal low pass filter of
radius 30

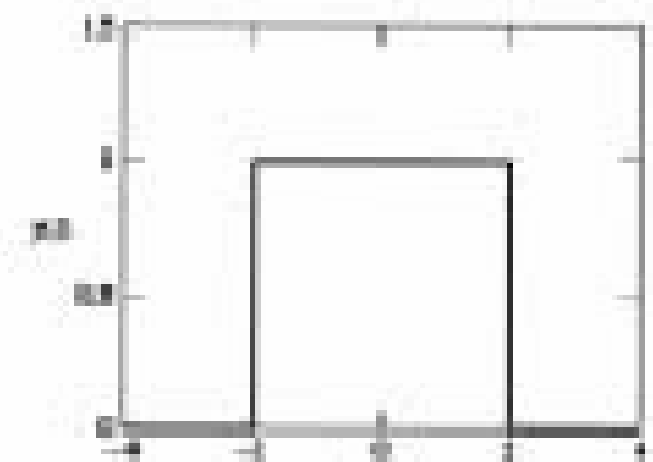
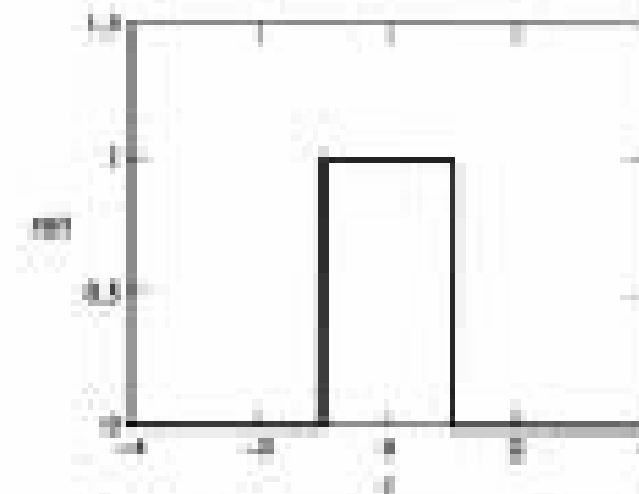
Result of filtering
with ideal low pass
filter of radius 80



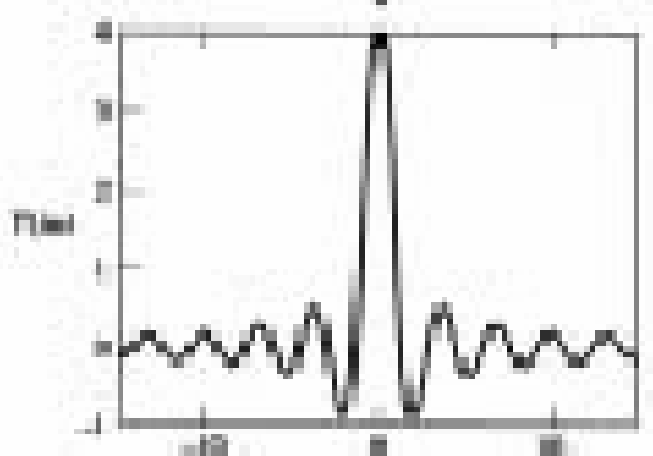
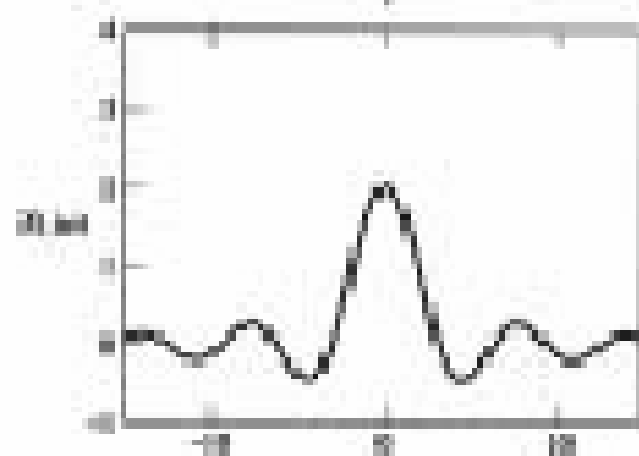
Result of filtering with
ideal low pass filter of
radius 230

"Ringing"

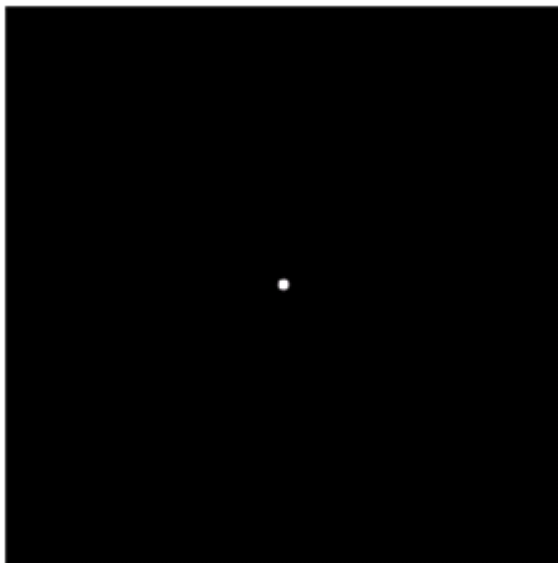
Spatial domain



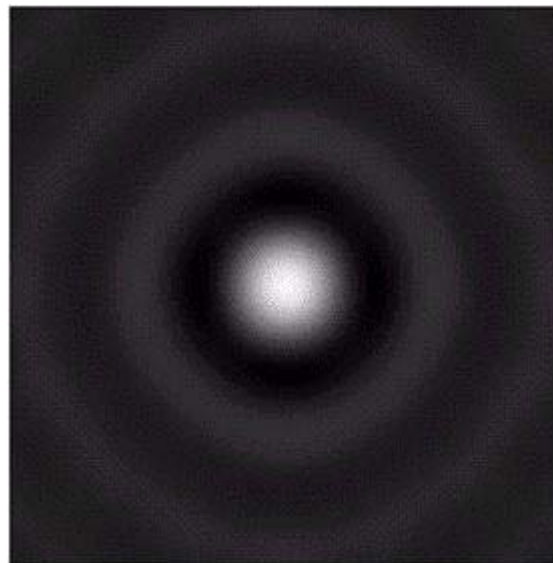
Frequency domain



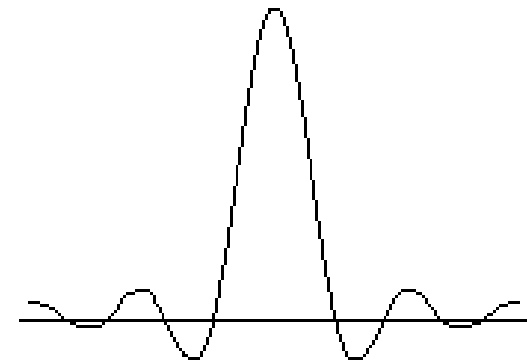
```
f = imread('bubbles.tif'); % only to get dimensions  
[M N] = size(f);  
D0 = 12; Hilp= lpfilter('ideal', M, N, D0);  
figure; imshow(fftshift(Hilp),[]);  
hilp= ifft2(Hilp);  
figure; imshow(ifftshift(abs(hilp)), []);  
figure; imshow(log(1 + ifftshift(abs(hilp)))), []);
```



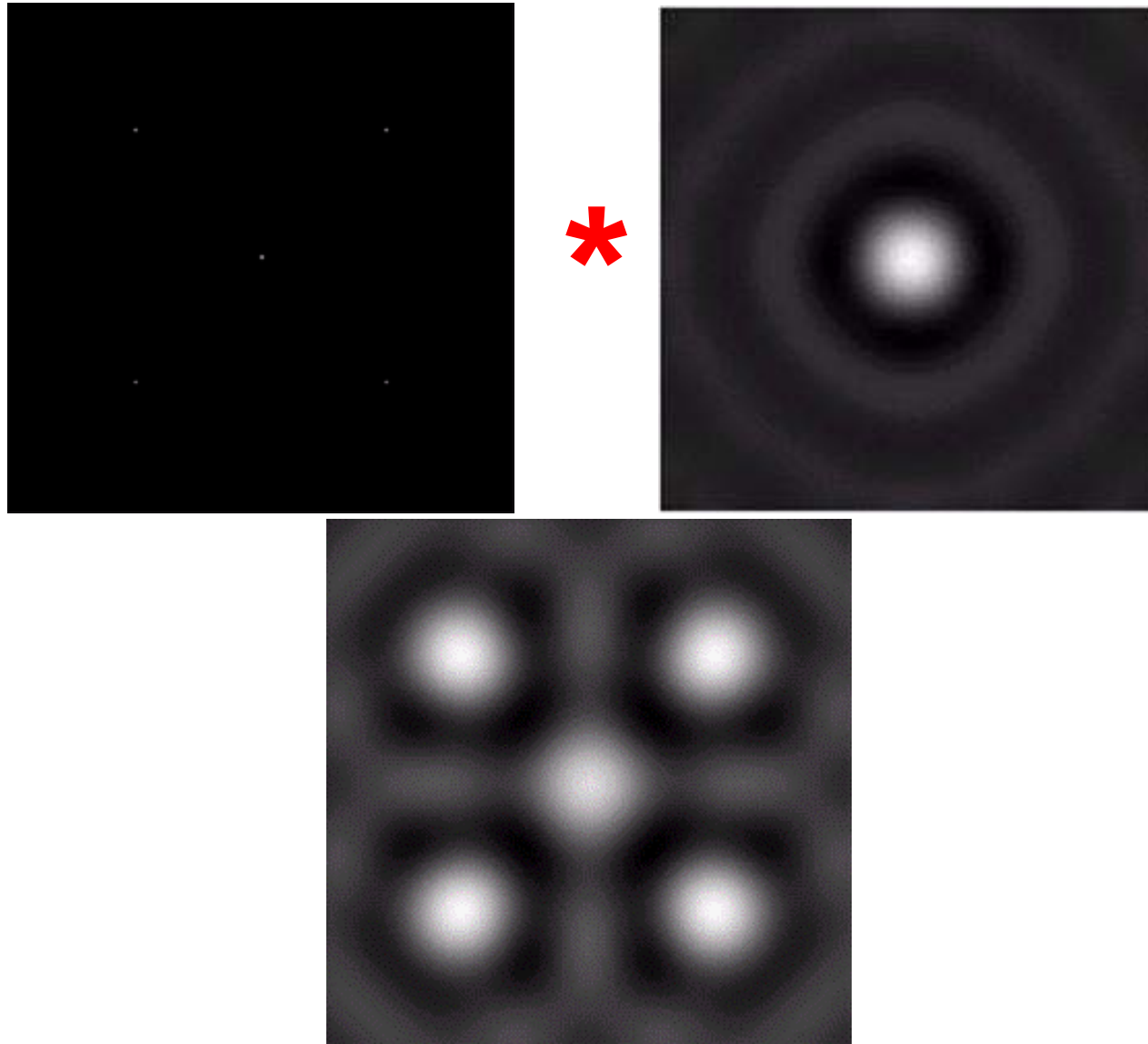
Frequency domain



Spatial domain



Convolution Theorem



Smoothing Frequency Domain Filters—Butterworth Low pass

The transfer function of a Butterworth lowpass filter of order n with cutoff frequency at distance D_0 from the origin is defined as:

$$H(u, v) = \frac{1}{1 + [D(u, v) / D_0]^{2n}}$$

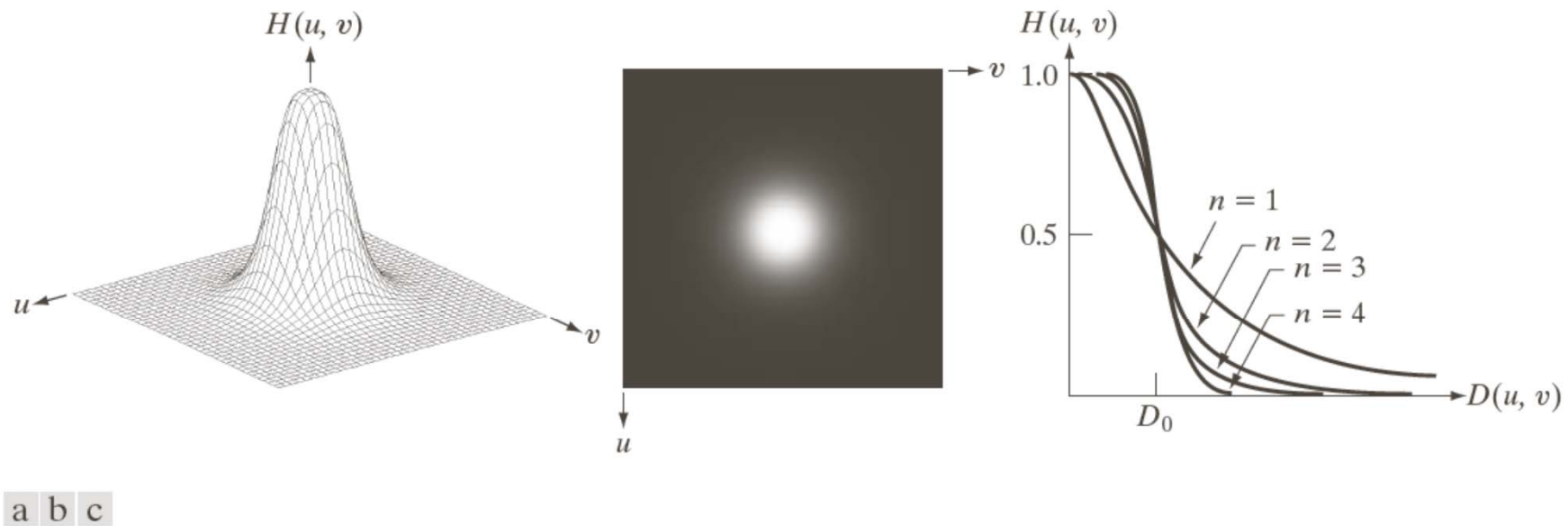


FIGURE 4.44 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

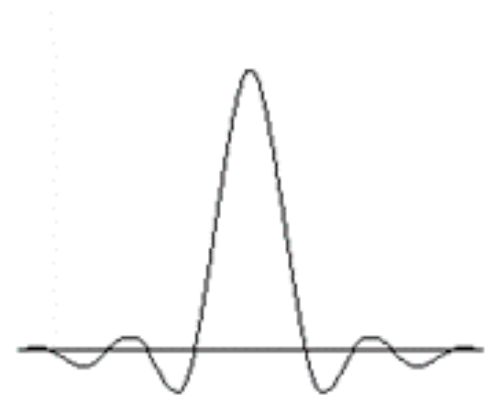
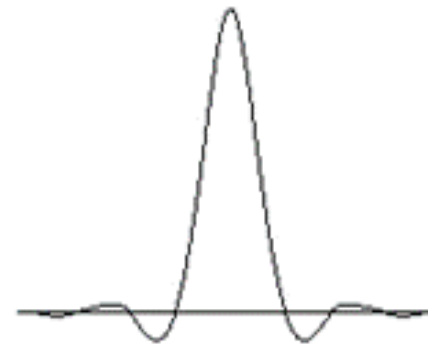
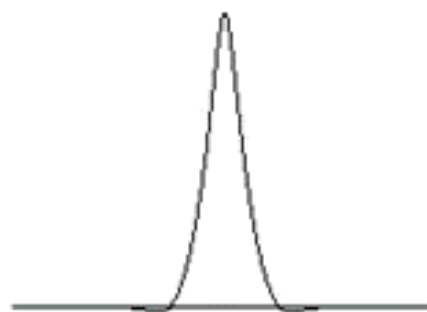
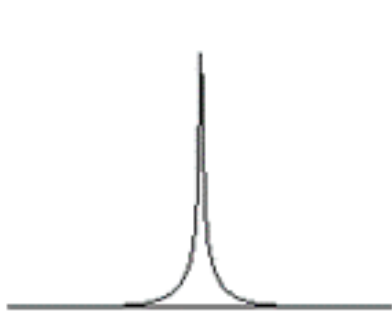
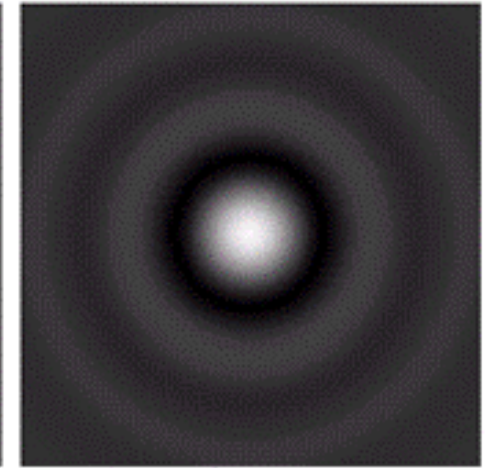
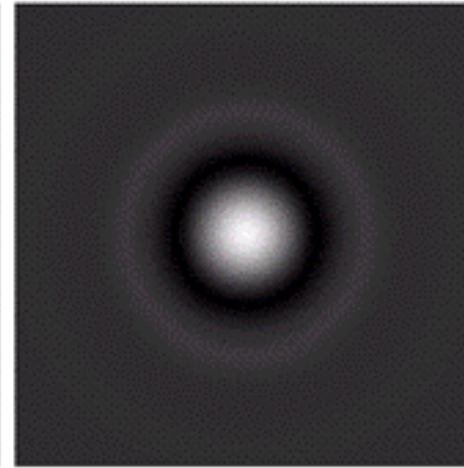
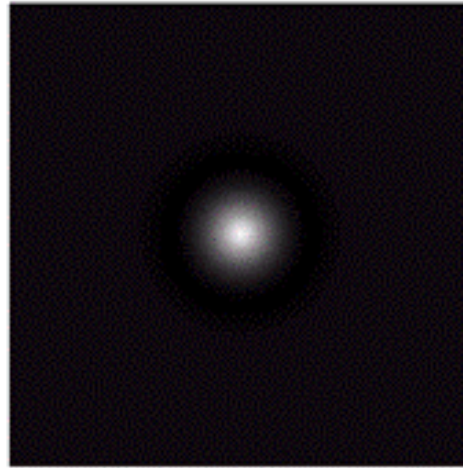
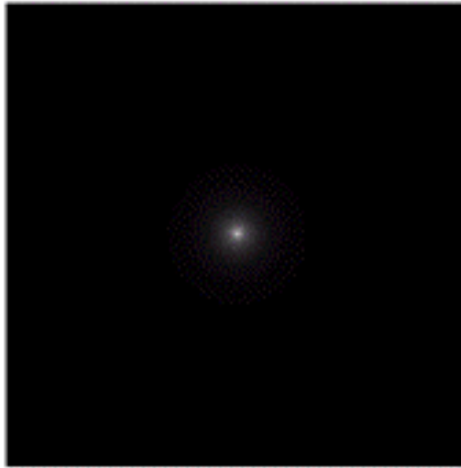
Frequency

Order 1

Order 2

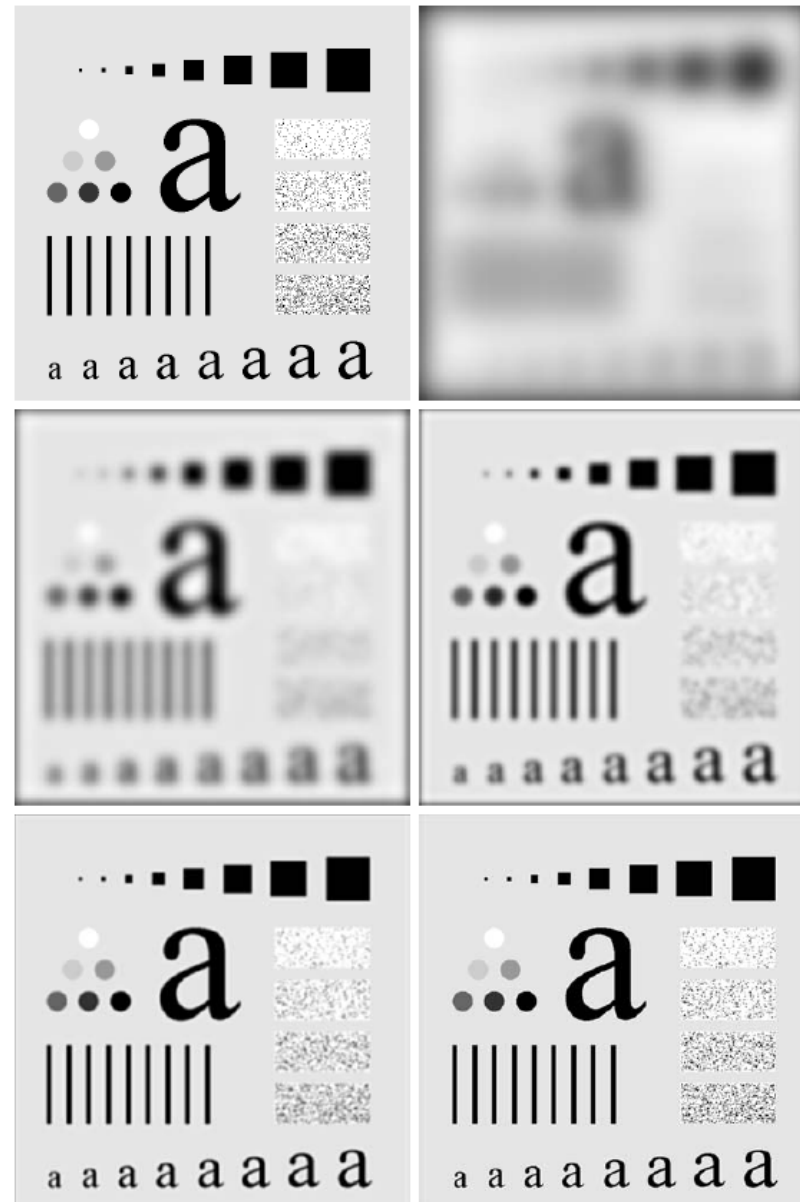
Order 5

Order 20



Spatial

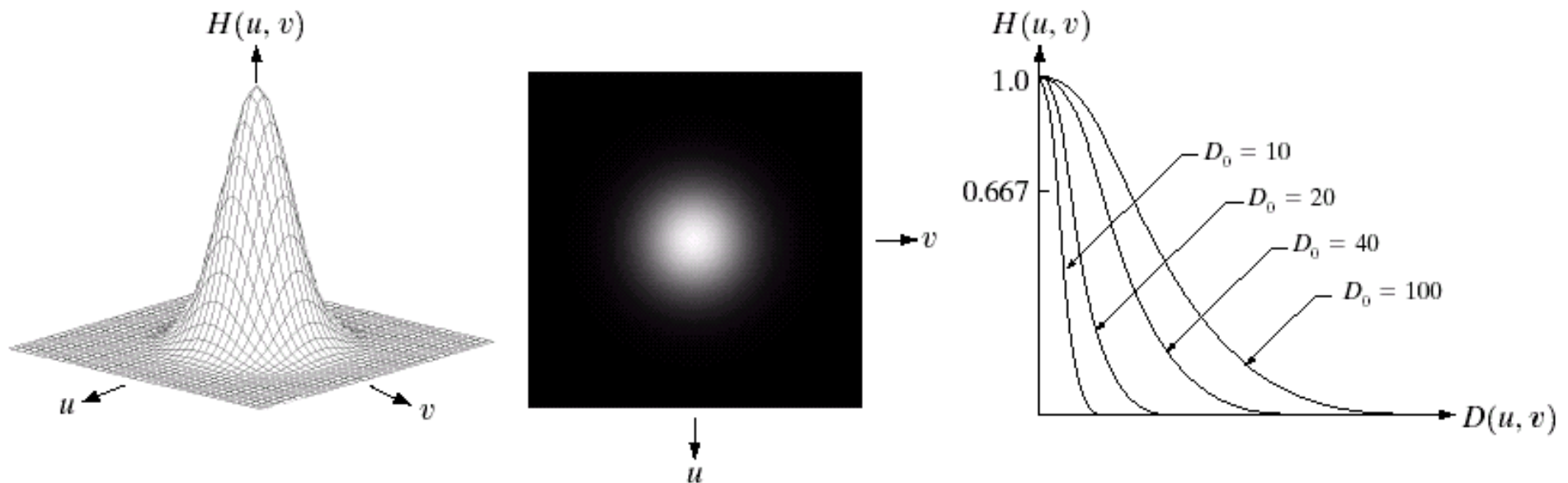
Smoothing Frequency Domain Filters—Butterworth Low pass



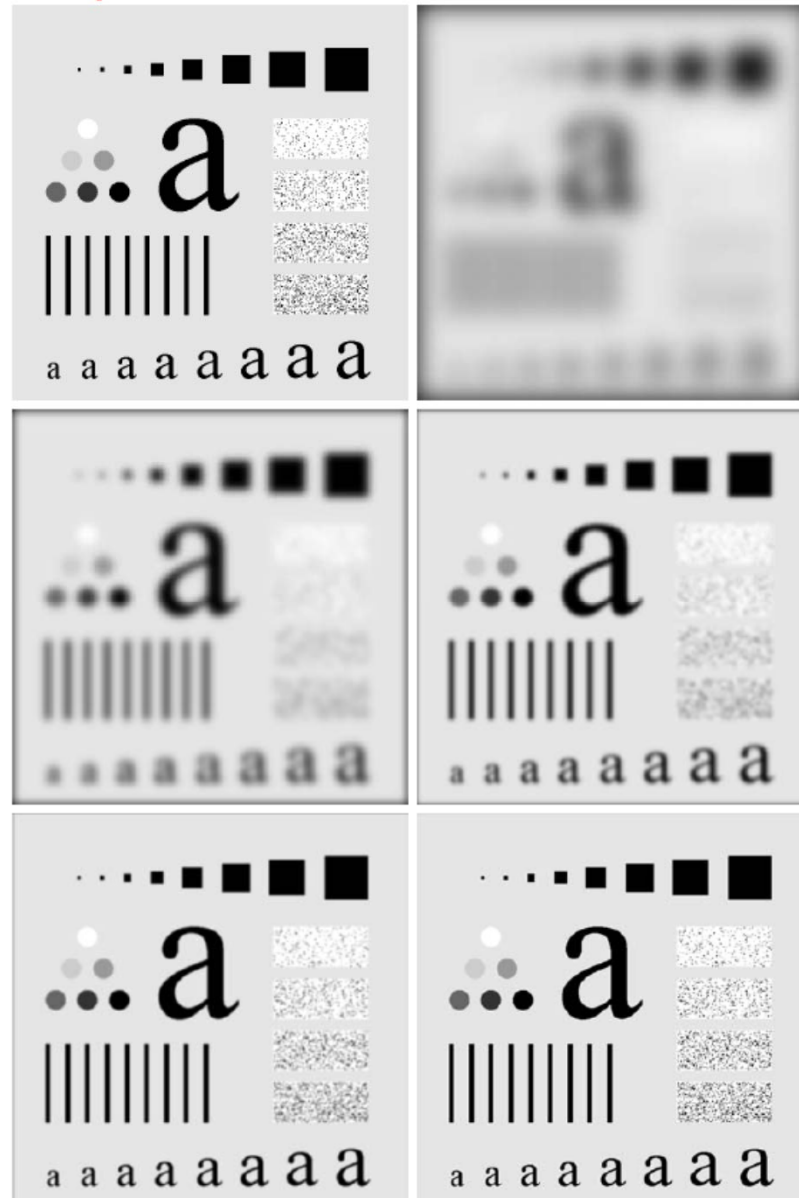
Smoothing Frequency Domain Filters—Gaussian Low pass

The transfer function of a Gaussian lowpass filter is defined as:

$$H(u, v) = e^{-D^2(u, v) / 2D_0^2}$$

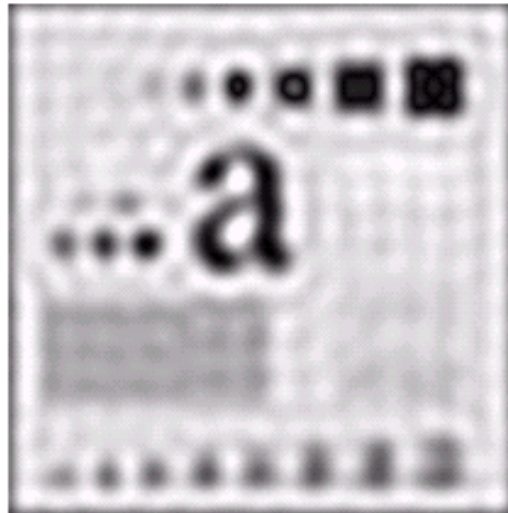


Smoothing Frequency Domain Filters—Gaussian Low pass

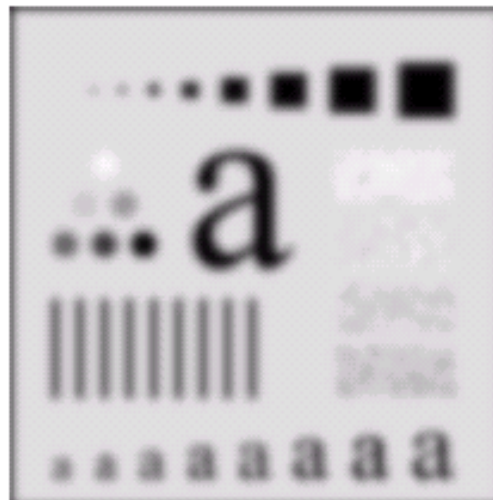
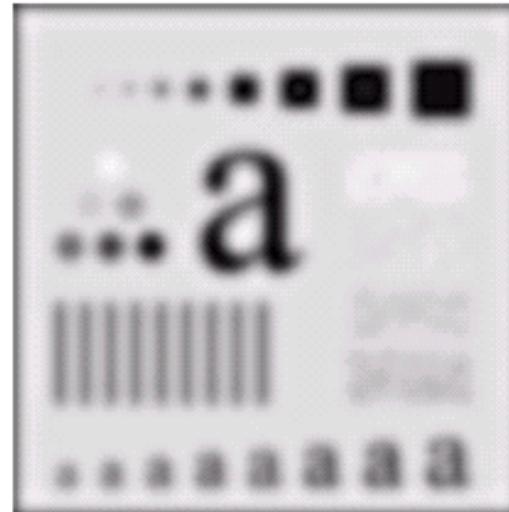


Lowpass Filters Compared

Result of
filtering with
ideal low pass
filter of radius
15



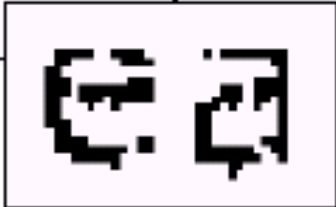
Result of filtering
with Butterworth
filter of order 2 and
cutoff radius 15



Result of filtering with Gaussian filter with cutoff radius 15

Smoothing Frequency Domain Filters—Gaussian Low pass

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



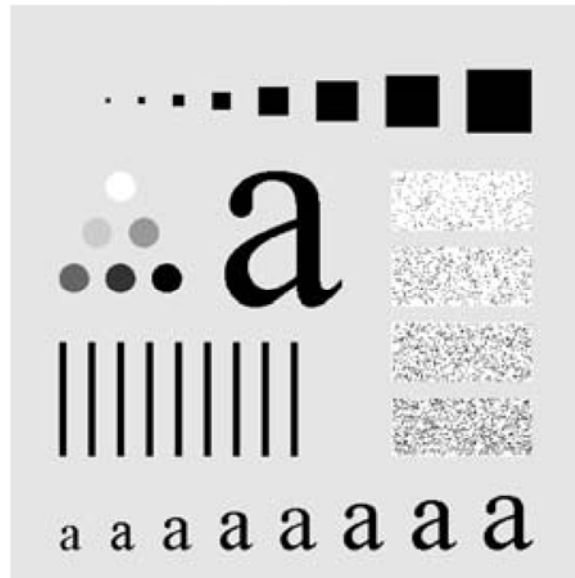
ea

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

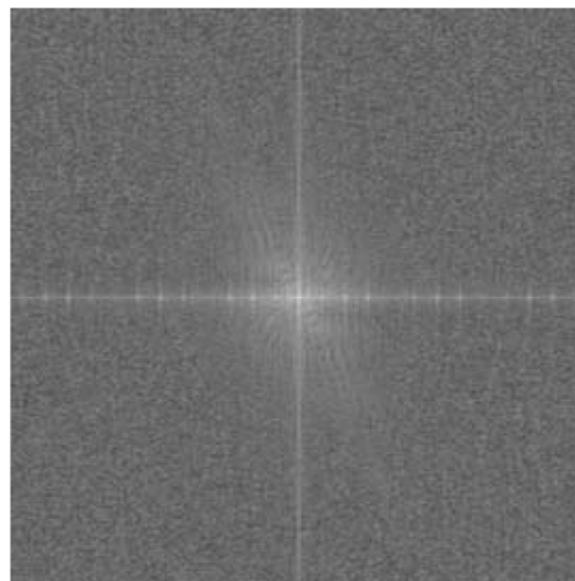
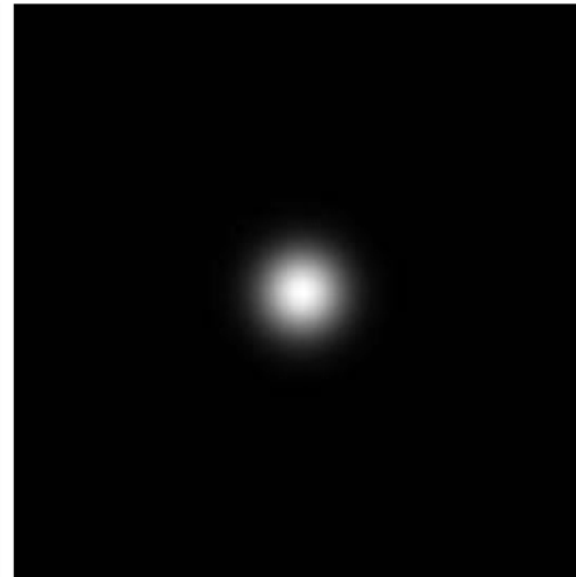


ea

Original image



Gaussian lowpass filter

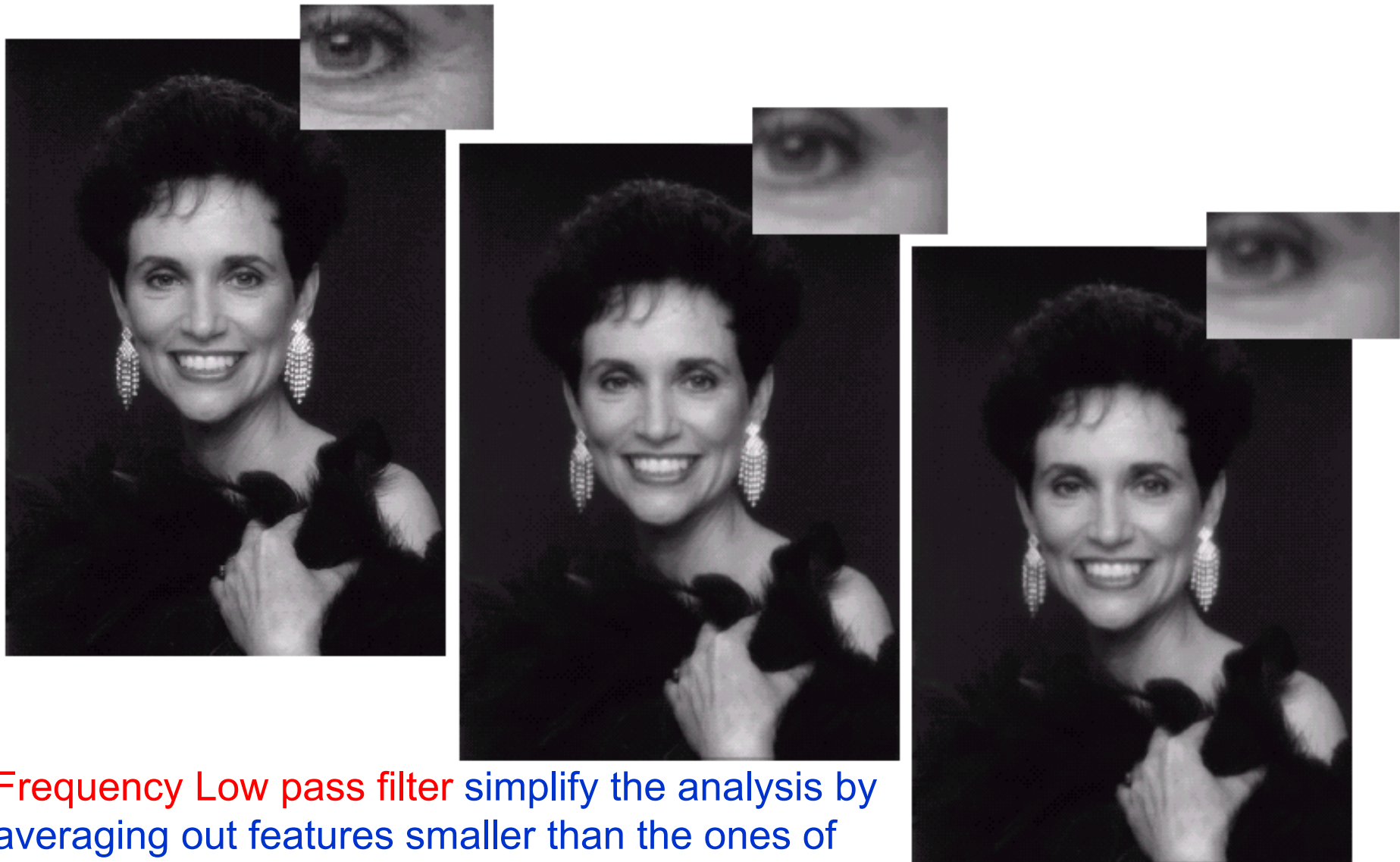


Spectrum of original image



Processed image

Smoothing Frequency Domain Filters—Gaussian Low pass



Frequency Low pass filter simplify the analysis by averaging out features smaller than the ones of interest.

Sharpening in the Frequency Domain

Edges and fine detail in images are associated with high frequency components

High pass filters – only pass the high frequencies, reverse of low pass filters, so:

$$H_{hp}(u, v) = 1 - H_{lp}(u, v)$$

The **ideal high pass** filter is given as:

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

where D_0 is the cut off distance as before

Sharpening in the Frequency Domain

The **Butterworth** high pass filter is given as:

$$H(u, v) = \frac{1}{1 + [D_0 / D(u, v)]^{2n}}$$

where n is the order and D_0 is the cut off distance as before

The **Gaussian** high pass filter is given as:

$$H(u, v) = 1 - e^{-D^2(u, v) / 2D_0^2}$$

where D_0 is the cut off distance as before

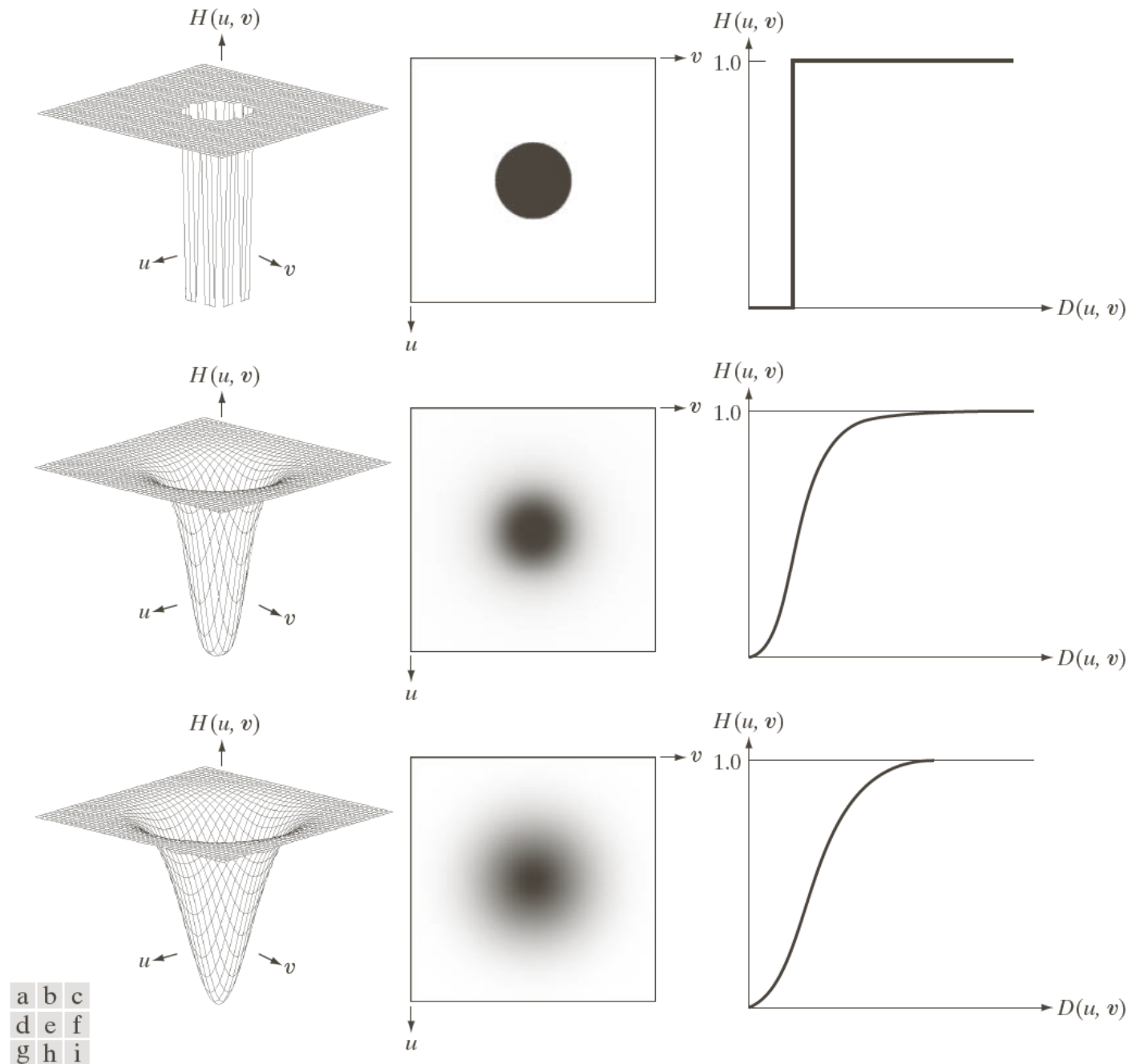


FIGURE 4.52 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

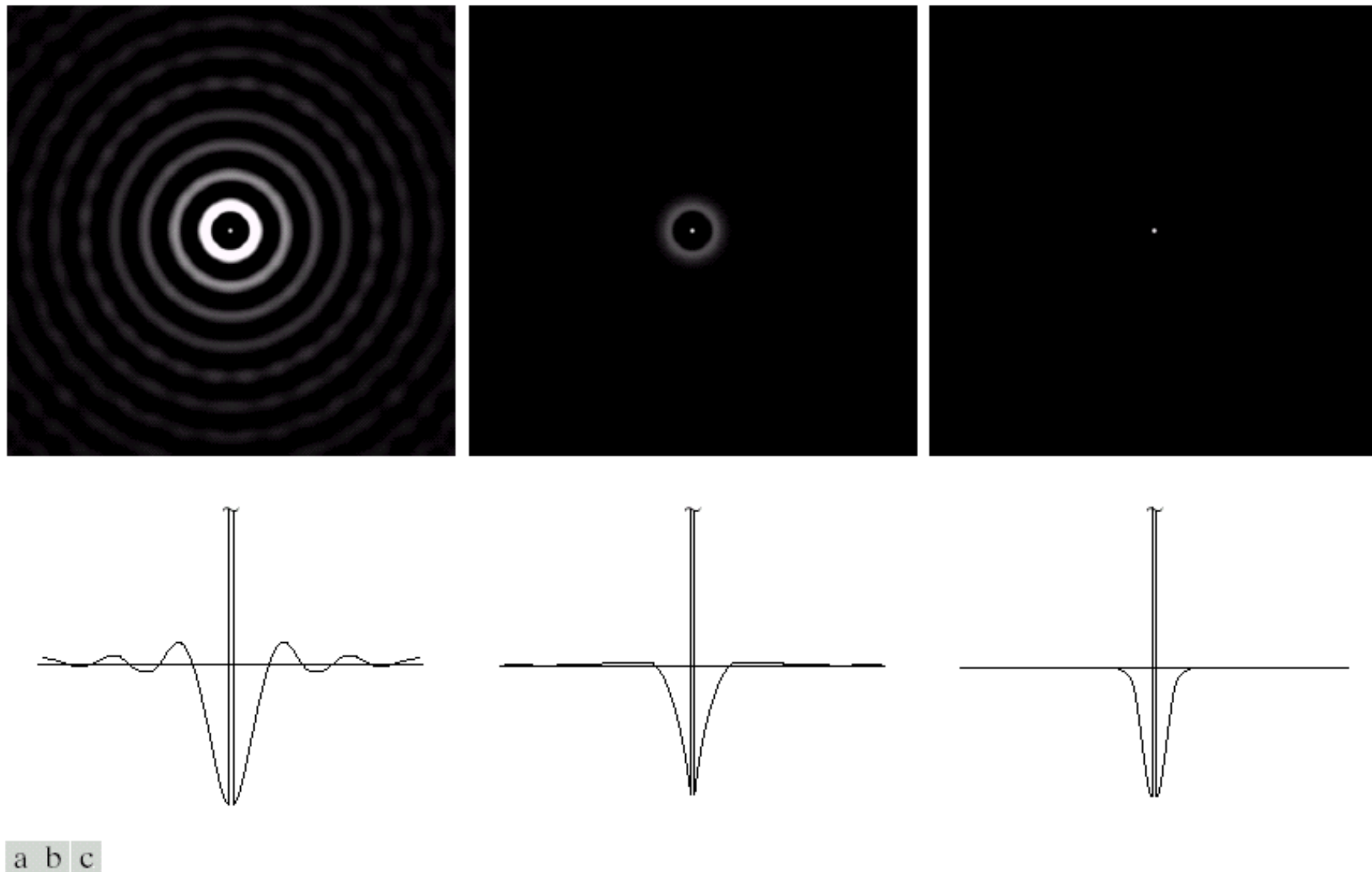
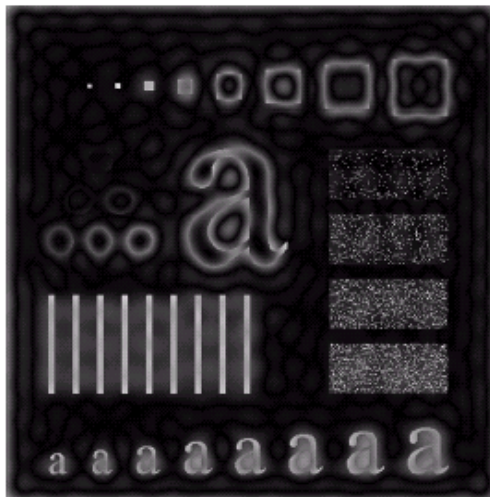
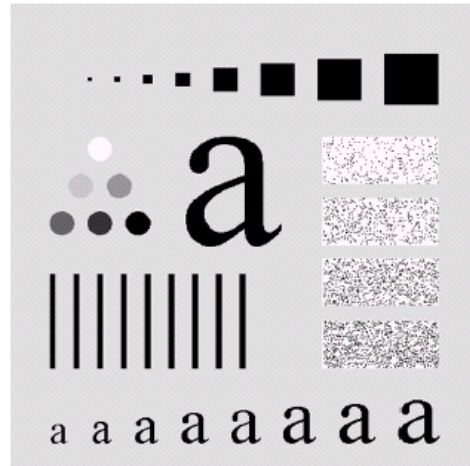


FIGURE 4.23 Spatial representations of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding gray-level profiles.

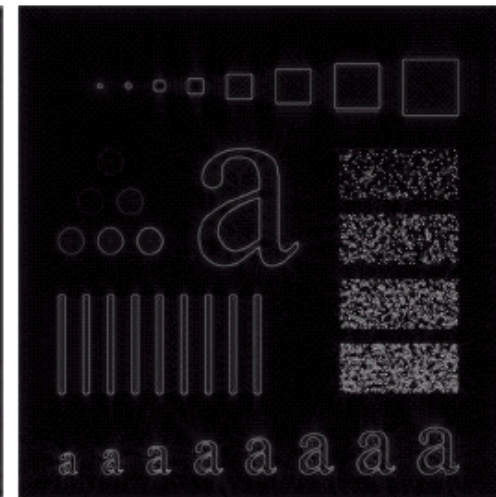
Sharpening in the Frequency Domain-Ideal high-pass



Results of ideal high pass
filtering with $D_0 = 15$

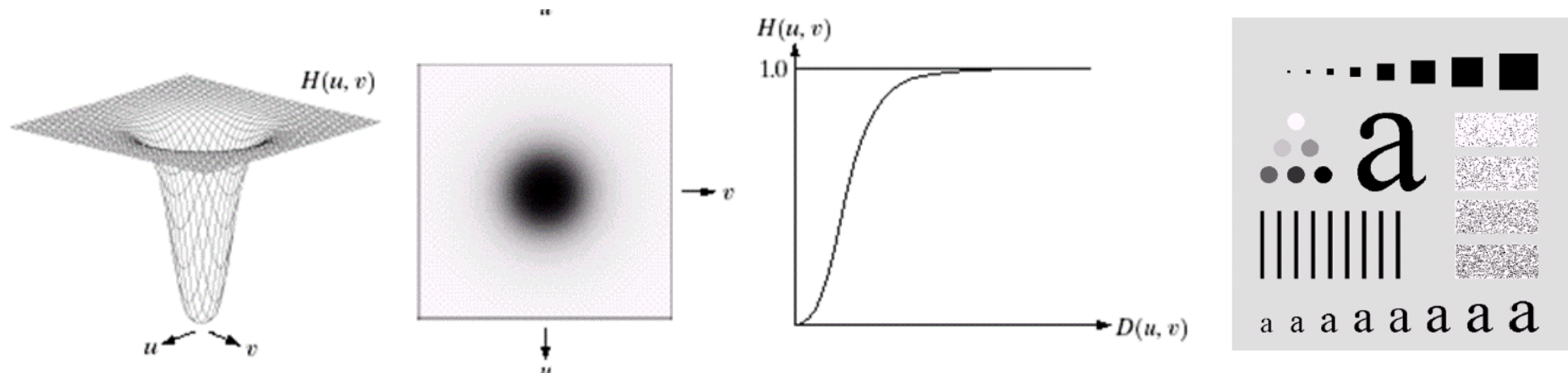


Results of ideal high pass
filtering with $D_0 = 30$

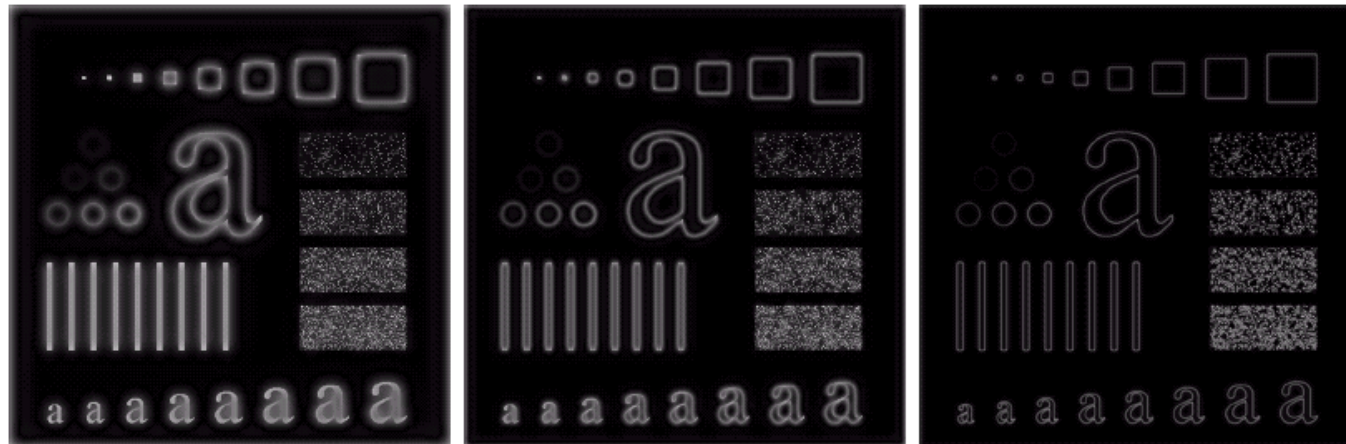


Results of ideal high pass
filtering with $D_0 = 80$

Sharpening -butterworth highpass



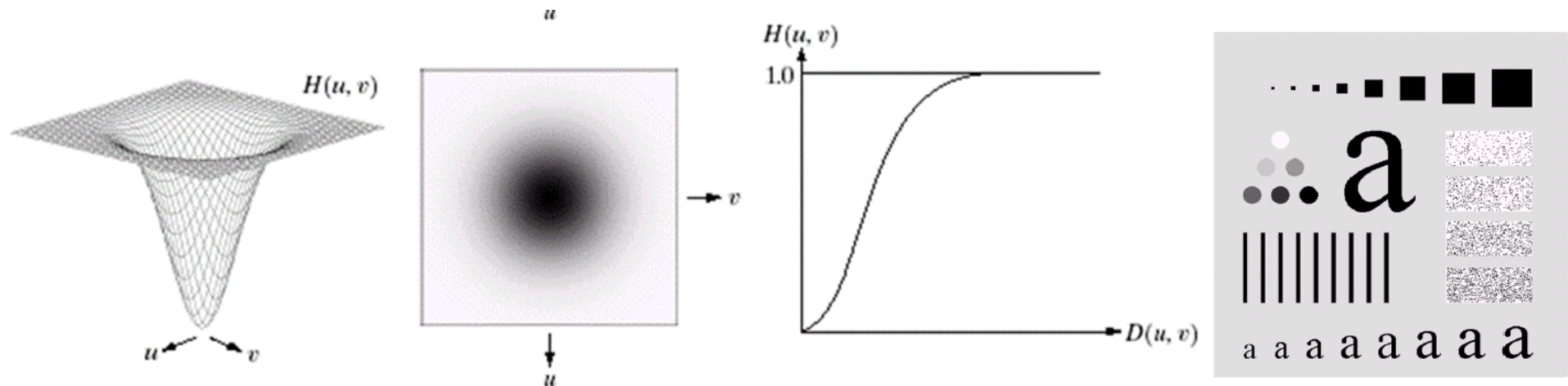
Butterworth
high pass
filtering of
order 2
with $D_0 =$
15



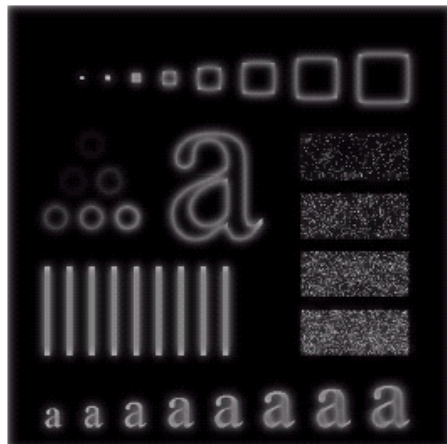
Results of Butterworth high pass filtering of
order 2 with $D_0 = 30$

Butterworth
high pass
filtering of
order 2
with $D_0 =$
80

Sharpening- Gaussian highpass



Gaussian
high pass
filtering
with $D_0 = 15$



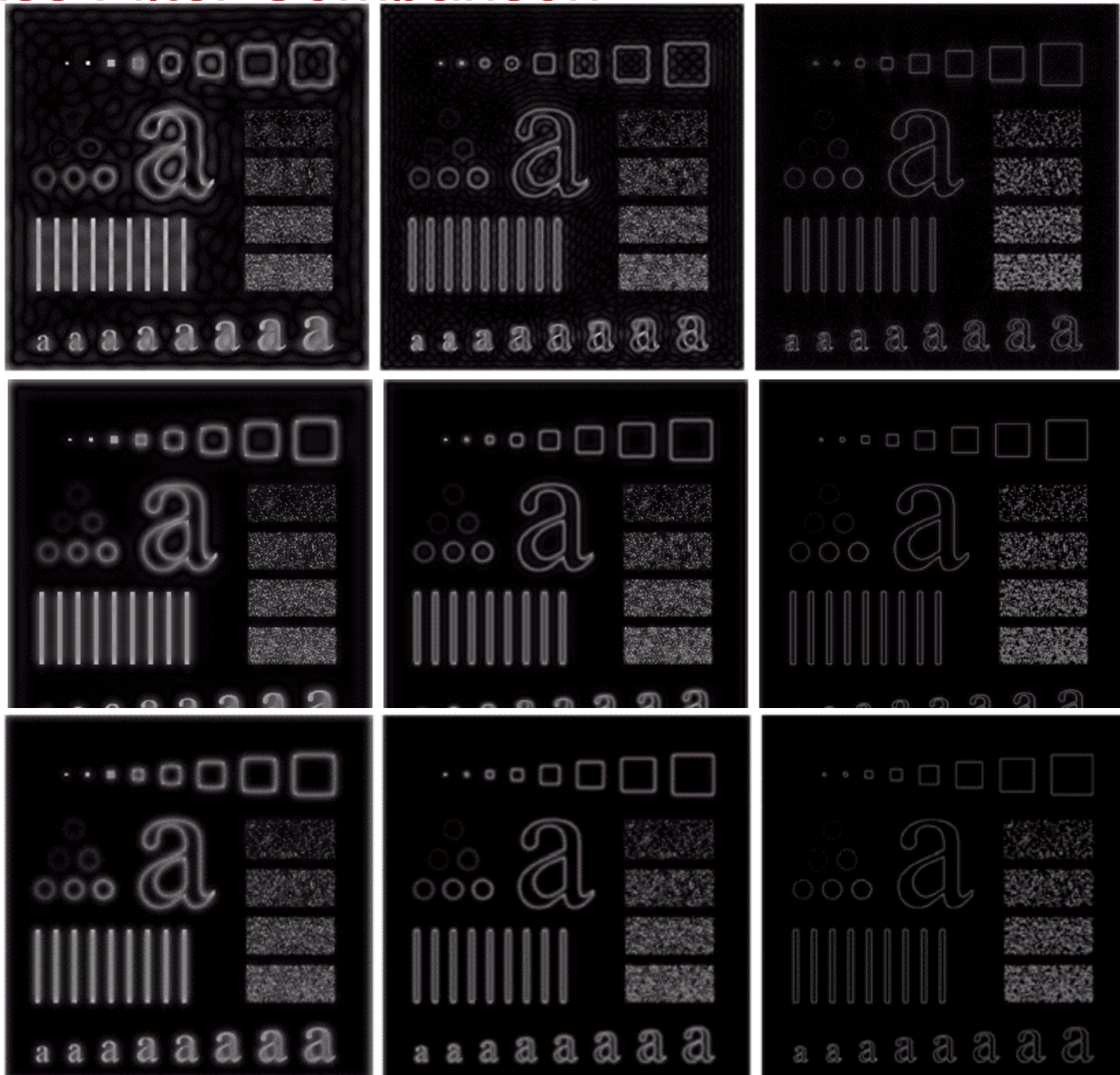
Results of Gaussian high
pass filtering with $D_0 = 30$



Gaussian
high pass
filtering
with $D_0 = 80$

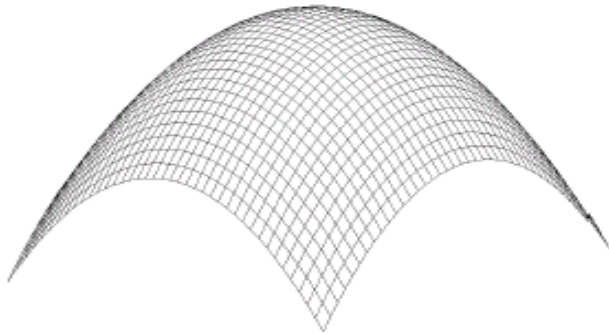


Highpass Filter Comparison

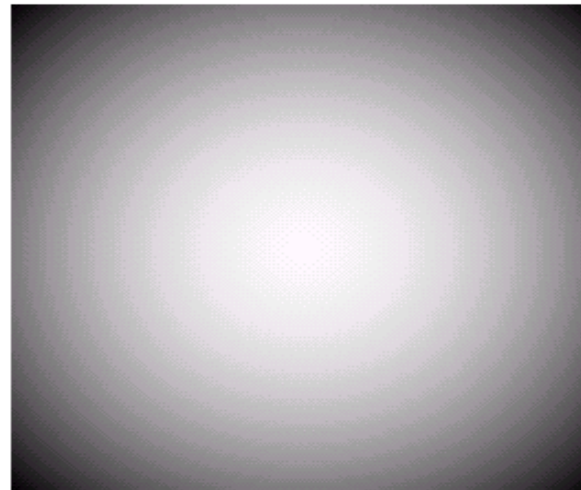


Laplacian In The Frequency Domain

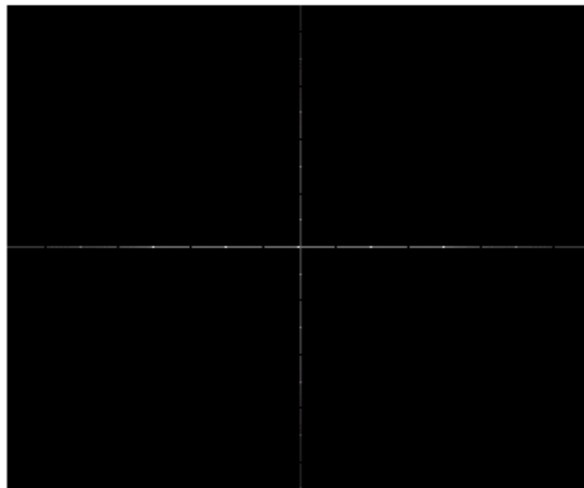
Laplacian in the frequency domain



2-D image of Laplacian in the frequency domain



Inverse DFT of
Laplacian in the
frequency domain



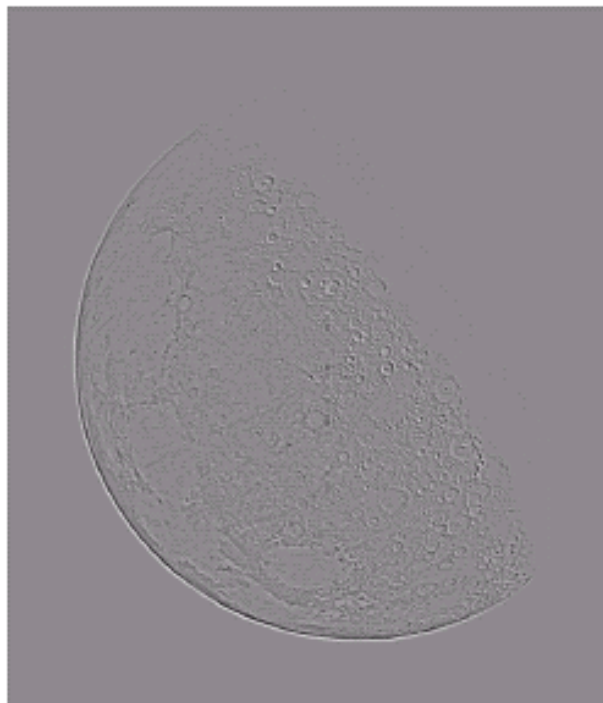
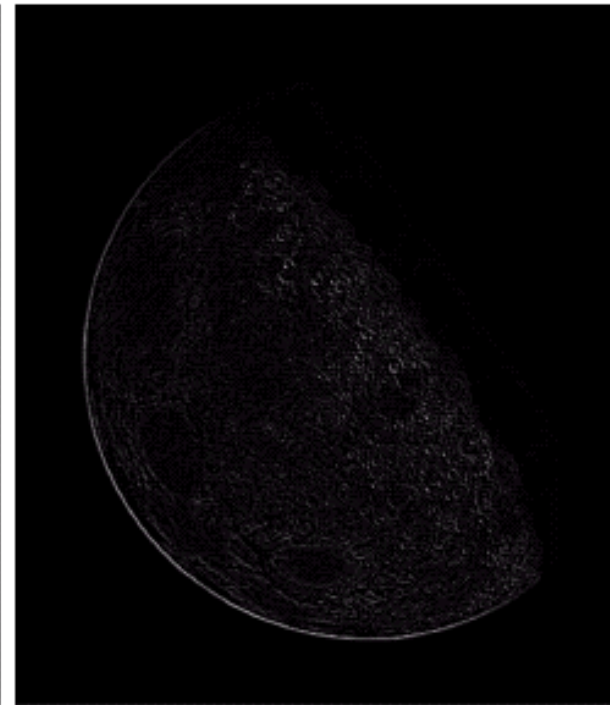
0	1	0
1	-4	1
0	1	0

Zoomed section
of the image on
the left compared
to spatial filter



a b c

FIGURE 4.57 (a) Thumb print. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)



Fast Fourier Transform

Allows the Fourier transform to be carried out in a reasonable amount of time

Reduces the amount of time required to perform a Fourier transform by a factor of 100 – 600 times!

